

Red Blood Cell and Bleeding Disorders

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This chapter first considers diseases of red cells. By far, the most common and clinically important are the anemias, red cell deficiency states that usually have a nonneoplastic basis. We will then complete our review of blood diseases by discussing the major bleeding disorders and complications of blood transfusion.

ANEMIAS

Anemia is defined as a reduction of the total circulating red cell mass below normal limits. Anemia reduces the oxygen-carrying capacity of the blood, leading to tissue hypoxia. In practice, the measurement of red cell mass is not easy, and anemia is usually diagnosed based on reductions in the *hematocrit* (the ratio of packed red cells to total blood volume) and the *hemoglobin concentration* to levels in the blood that are below the normal range. These values correlate with the red cell mass except when there are changes in plasma volume caused by fluid retention or dehydration.

There are many classifications of anemia. The one described in this chapter is based on underlying mechanisms (Table 14.1). A second clinically useful approach classifies anemia according to alterations in red cell morphology, which often point to specific causes. Morphologic characteristics that provide etiologic clues include red cell size

(normocytic, microcytic, or macrocytic); degree of hemoglobinization, reflected in the color of red cells (normochromic or hypochromic); and shape. Microcytic hypochromic anemias are caused by disorders of hemoglobin synthesis, and macrocytic anemias often stem from abnormalities that impair the maturation of erythroid precursors in the bone marrow. Normochromic, normocytic anemias have diverse etiologies; in some of these anemias, characteristic abnormalities of red cell shape provide an important clue as to the cause. Red cell shape is assessed through visual inspection of peripheral smears, whereas as other red cell indices are determined in clinical laboratories with special instrumentation. The most useful of these indices are:

- *Mean cell volume*: the average volume of a red cell expressed in femtoliters (fL)
- *Mean cell hemoglobin*: the average content (mass) of hemoglobin per red cell, expressed in picograms (pg)
- *Mean cell hemoglobin concentration*: the average concentration of hemoglobin in a given volume of packed red cells, expressed in grams per deciliter (g/dL)
- *Red cell distribution width*: the coefficient of variation of red cell volume

Adult reference ranges for red cell indices are shown in Table 14.2.

TABLE 14.1 Classification of Anemia According to Underlying Mechanism

Mechanism	Specific Examples
Blood Loss	
Acute blood loss	Trauma
Chronic blood loss	Gastrointestinal tract lesions, gynecologic disturbances ^a
Increased Red Cell Destruction (Hemolysis)	
Inherited Genetic Defects	
Red cell membrane disorders	Hereditary spherocytosis, hereditary elliptocytosis
Enzyme deficiencies	
Hexose monophosphate shunt enzyme deficiencies	G6PD deficiency, glutathione synthetase deficiency
Glycolytic enzyme deficiencies	Pyruvate kinase deficiency, hexokinase deficiency
Hemoglobin Abnormalities	
Deficient globin synthesis	Thalassemia syndromes
Structurally abnormal globins (hemoglobinopathies)	Sickle cell disease, unstable hemoglobins
Acquired Genetic Defects	
Deficiency of phosphatidylinositol-linked glycoproteins	Paroxysmal nocturnal hemoglobinuria
Antibody-mediated destruction	Hemolytic disease of the newborn (Rh disease), transfusion reactions, drug-induced, autoimmune disorders
Mechanical Trauma	
Microangiopathic hemolytic anemias	Hemolytic uremic syndrome, disseminated intravascular coagulation, thrombotic thrombocytopenia purpura
Cardiac traumatic hemolysis	Defective cardiac valves
Repetitive physical trauma	Bongo drumming, marathon running, karate chopping
Infections of Red Cells	
	Malaria, babesiosis
Toxic or Chemical Injury	
	Clostridial sepsis, snake venom, lead poisoning
Membrane Lipid Abnormalities	
	Abetalipoproteinemia, severe hepatocellular liver disease
Sequestration	
	Hypersplenism
Decreased Red Cell Production	
Inherited Genetic Defects	
Defects leading to stem cell depletion	Fanconi anemia, telomerase defects
Defects affecting erythroblast maturation	Thalassemia syndromes
Nutritional Deficiencies	
Deficiencies affecting DNA synthesis	B ₁₂ and folate deficiencies
Deficiencies affecting hemoglobin synthesis	Iron deficiency
Erythropoietin Deficiency	
	Renal failure, anemia of chronic inflammation
Immune-Mediated Injury of Progenitors	
	Aplastic anemia, pure red cell aplasia
Inflammation-Mediated Iron Sequestration	
	Anemia of chronic inflammation
Primary Hematopoietic Neoplasms	
	Acute leukemia, myelodysplastic syndrome, myeloproliferative neoplasms (Chapter 13)
Space-Occupying Marrow Lesions	
	Metastatic neoplasms, granulomatous disease
Infections of Red Cell Progenitors	
	Parvovirus B19 infection
Unknown Mechanisms	
	Endocrine disorders, hepatocellular liver disease

^aMost often anemia stems from iron deficiency, not bleeding per se.
G6PD, Glucose-6-phosphate dehydrogenase.

Whatever its cause, when sufficiently severe, anemia leads to manifestations related to the diminished ability of the blood to deliver oxygen to the tissues. Patients often report weakness, malaise, easy fatigability, and dyspnea on mild exertion. Hypoxia may cause fatty change in the liver, myocardium, and kidney. On occasion, myocardial hypoxia manifests as angina pectoris, particularly when complicated by preexisting coronary artery disease. Central nervous system hypoxia may cause headache, dimness of vision, and faintness. With

acute blood loss and shock, oliguria and anuria may develop due to renal hypoperfusion.

Anemias of Blood Loss

Acute Blood Loss

The effects of acute blood loss are mainly due to the loss of intravascular volume, which, if massive, can lead to cardiovascular collapse,

TABLE 14.2 Adult Reference Ranges for Red Cells^a

Measurement	Males	Females
Hemoglobin (g/dL)	13.6–17.2	12.0–15.0
Hematocrit (%)	39–49	33–43
Red cell count ($\times 10^6/\mu\text{L}$)	4.3–5.9	3.5–5.0
Reticulocyte count (%)	0.5–1.5	
Mean cell volume (fL)	82–96	
Mean cell hemoglobin (pg)	27–33	
Mean cell hemoglobin concentration (g/dL)	33–37	
Red cell distribution width	11.5–14.5	

^aReference ranges vary among laboratories. The reference ranges for the laboratory providing the result should always be used in interpreting test results.

shock, and death. The clinical features depend on the rate of hemorrhage and whether the bleeding is external or internal. If the patient survives, the blood volume is rapidly restored by movement of water from the interstitial fluid compartment to the intravascular compartment. This fluid shift produces hemodilution and lowers the hematocrit. The resulting reduction in tissue oxygenation is sensed by specialized stromal cells in the kidney that respond by releasing erythropoietin, stimulating the proliferation of committed erythroid progenitors (colony-forming unit–erythroid [CFU-E]) in the marrow (see Fig. 13.1). It takes about 5 days for the progeny of these CFU-Es to mature and appear as newly released red cell precursors (*reticulocytes*) in the peripheral blood. The iron in hemoglobin is recaptured if red cells extravasate into tissues, whereas bleeding into the gut or out of the body leads to iron loss and possible iron deficiency, which can hamper the restoration of normal red cell counts.

Significant bleeding results in predictable changes in the blood involving not only red cells, but also white cells and platelets. If the bleeding is sufficiently massive to cause a decrease in blood pressure, the compensatory release of adrenergic hormones mobilizes granulocytes from the intravascular marginal pool and results in leukocytosis. Initially, red cells appear normal in size and color (normocytic, normochromic). However, as marrow production increases, there is a striking increase in the reticulocyte count (reticulocytosis), which reaches 10% to 15% after 7 days. Reticulocytes are larger than normal red cells and have blue-red polychromatophilic cytoplasm due to the presence of RNA, a feature that allows them to be identified in the clinical laboratory. Early recovery from blood loss is also often accompanied by thrombocytosis, which results from an increase in platelet production.

Chronic Blood Loss

Chronic blood loss induces anemia when the rate of loss exceeds the regenerative capacity of the marrow or when iron reserves are depleted and iron deficiency anemia appears (see later).

Hemolytic Anemias

Hemolytic anemias are the result of extravascular or intravascular hemolysis. They share the following features:

- A shortened red cell life span below the normal 120 days
- Elevated erythropoietin levels and a compensatory increase in erythropoiesis
- Accumulation of hemoglobin degradation products that are created during the process of red cell hemolysis

The physiologic destruction of senescent red cells takes place within macrophages, which are abundant in the spleen, liver, and bone

marrow. This process appears to be triggered by age-dependent changes in red cell surface proteins, which lead to their recognition and removal by phagocytes. In the great majority of hemolytic anemias, the premature destruction of red cells also occurs within phagocytes, an event that is referred to as *extravascular hemolysis*. If persistent, extravascular hemolysis leads to hyperplasia of phagocytes manifested by varying degrees of splenomegaly.

Extravascular hemolysis is usually caused by alterations that make red cells less deformable. Extreme changes in shape are required for red cells to navigate from the splenic cords to the splenic sinusoids successfully. Reduced deformability makes this passage difficult, leading to red cell sequestration and phagocytosis by macrophages located within the cords. Regardless of the cause, the principal clinical features of extravascular hemolysis are anemia, splenomegaly, and jaundice. Some hemoglobin inevitably escapes from phagocytes, which leads to variable decreases in plasma *haptoglobin*, an α_2 -globulin that binds free hemoglobin and prevents its excretion in the urine. Because much of the premature destruction of red cells occurs in the spleen, individuals with extravascular hemolysis often benefit from splenectomy.

Intravascular hemolysis of red cells may be caused by mechanical injury, complement fixation, intracellular parasites (e.g., falciparum malaria, Chapter 8), or exogenous toxic factors. It is less common than extravascular hemolysis. Sources of mechanical injury include trauma caused by cardiac valves, narrowing of the microcirculation by thrombi or tumor cells, or repetitive physical trauma (e.g., marathon running and bongo drum beating). Complement fixation occurs in a variety of situations in which antibodies recognize and bind red cell antigens. Toxic injury is exemplified by clostridial sepsis, which results in the release of enzymes that digest the red cell membrane.

Whatever the mechanism, intravascular hemolysis is manifested by anemia, hemoglobinemia, hemoglobinuria, hemosiderinuria, and jaundice. Free hemoglobin released from lysed red cells is promptly bound by haptoglobin, producing a complex that is rapidly cleared by mononuclear phagocytes. As serum haptoglobin is depleted, free hemoglobin oxidizes to methemoglobin, which is brown in color. The renal proximal tubular cells reabsorb and break down much of the filtered hemoglobin and methemoglobin, but some passes out in the urine, imparting a red-brown color. Iron released from hemoglobin can accumulate within tubular cells, giving rise to *renal hemosiderosis*. If the hemolysis is chronic, iron loss in the urine and from sloughed renal tubular cells may lead to iron deficiency. Concomitantly, heme groups derived from hemoglobin-haptoglobin complexes are metabolized to bilirubin within mononuclear phagocytes, leading to jaundice. Unlike in extravascular hemolysis, splenomegaly is not seen.

In all types of uncomplicated hemolytic anemia, the excess serum bilirubin is unconjugated. The level of hyperbilirubinemia depends on the functional capacity of the liver and the rate of hemolysis. When the liver is normal, jaundice is rarely severe, but excessive bilirubin excreted by the liver into the biliary tract often leads to the formation of gallstones derived from heme pigments.

MORPHOLOGY

Certain changes are seen in hemolytic anemia regardless of cause or type. Anemia and lowered tissue oxygen tension trigger the production of erythropoietin, which stimulates erythroid differentiation and leads to the appearance of **increased numbers of erythroid precursors (normoblasts)** in the marrow (Fig. 14.1). Compensatory increases in erythropoiesis result in a **prominent reticulocytosis** in the peripheral blood. The phagocytosis of red cells leads to the accumulation of the iron-containing pigment **hemosiderin**, particularly in the spleen, liver, and bone marrow. Such iron accumulation is referred to as

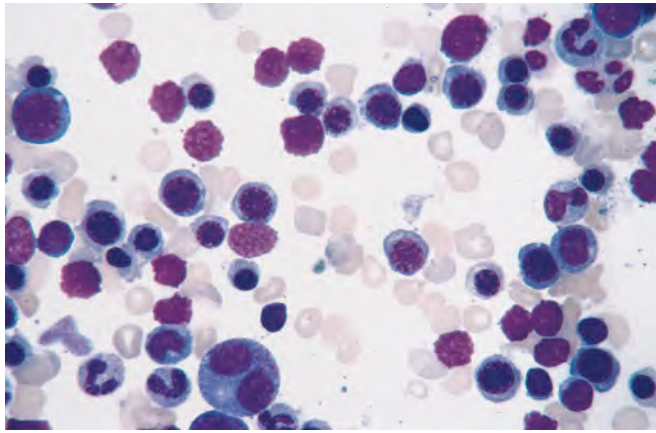


Fig. 14.1 Marrow aspirate smear from a patient with hemolytic anemia. There is an increased number of maturing erythroid progenitors (normoblasts). (Courtesy Dr. Steven Kroft, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

hemosiderosis. If the anemia is severe, **extramedullary hematopoiesis** can appear in the liver, spleen, and lymph nodes. With chronic hemolysis, elevated biliary excretion of bilirubin promotes the formation of **pigment gallstones** (cholelithiasis).

The hemolytic anemias can be classified in a variety of ways; here, we rely on the underlying mechanisms (see Table 14.1). We begin by discussing the major inherited forms of hemolytic anemia, and then move on to the acquired forms that are most common or of particular pathophysiologic interest.

Hereditary Spherocytosis

Hereditary spherocytosis (HS) is an inherited disorder caused by intrinsic defects in the red cell membrane skeleton that render red cells spheroid, less deformable, and vulnerable to splenic sequestration and destruction. The prevalence of HS is highest in northern Europe, where rates of 1 in 5000 are reported. An autosomal dominant inheritance pattern is seen in about 75% of cases. The remaining patients have a more severe form of the disease that is usually caused by the inheritance of two different defects (a state known as compound heterozygosity).

Pathogenesis

The remarkable deformability and durability of the normal red cell are attributable to the physicochemical properties of its specialized membrane skeleton (Fig. 14.2), which lies closely apposed to the internal surface of the plasma membrane. Its chief protein component, spectrin, consists of two polypeptide chains, α and β , which form intertwined (helical) flexible heterodimers. The “head” regions of spectrin dimers self-associate to form tetramers, and the “tails” associate with actin oligomers. Each actin oligomer can bind multiple spectrin tetramers, thus creating a two-dimensional spectrin-actin skeleton that is connected to the cell membrane by two distinct interactions. The first, involving the proteins ankyrin and band 4.2, binds spectrin to the transmembrane ion transporter, band 3. The second, involving protein 4.1, binds the “tail” of spectrin to another transmembrane protein, glycophorin A.

HS is caused by diverse mutations that lead to an insufficiency of membrane skeletal components (see Fig. 14.2). As a result of these alterations, the life span of affected red cells is decreased on average to

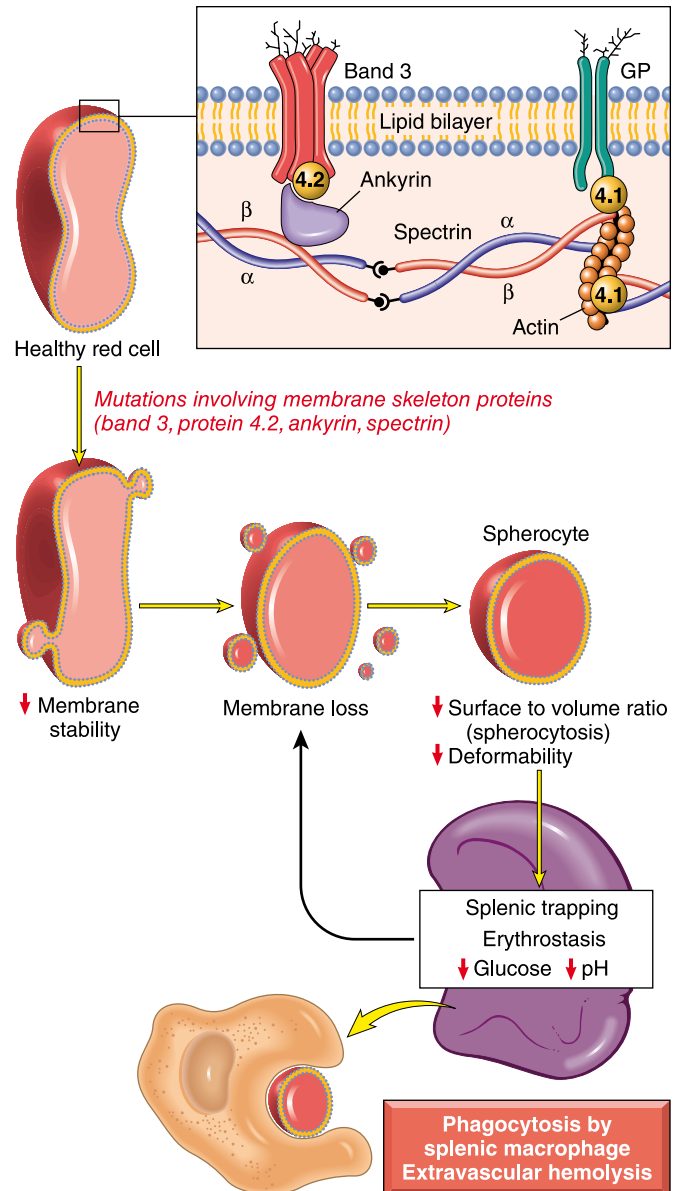


Fig. 14.2 Pathogenesis of hereditary spherocytosis. The upper panel shows the normal organization of the major red cell membrane skeletal proteins. Various mutations involving α -spectrin, β -spectrin, ankyrin, band 4.2, or band 3 that weaken the interactions between these proteins destabilize the red cell membrane and cause red cells to lose membrane fragments as they age. To accommodate the resultant change in the ratio of surface area to volume, these cells adopt a spherical shape. Spherocytic cells are less deformable than normal ones and therefore become trapped in the splenic cords, where they are phagocytosed by macrophages. GP, Glycophorin.

10 to 20 days from the normal 120 days. The pathogenic mutations most commonly affect ankyrin, band 3, spectrin, or band 4.2, proteins involved in one of the two tethering interactions. Most mutations cause frameshifts or introduce premature stop codons, such that the mutated allele fails to produce any protein. The resulting deficiency of the affected protein reduces assembly of the membrane skeleton as a whole, destabilizing the overlying plasma membrane. Young HS red cells are normal in shape, but the destabilized lipid bilayer sheds membrane fragments as the red cells age in the circulation. The loss of

membrane relative to cytoplasm “forces” the cells to assume the smallest possible diameter for a given volume, namely, a sphere. Compound heterozygosity for two defective alleles understandably results in more profound membrane skeleton deficiency and more severe disease.

The invariably beneficial effects of splenectomy prove that the spleen has a cardinal role in the premature demise of spherocytes. The travails of spherocytic red cells are well defined. In the life of the nondeformable spherocyte, the spleen is the villain (see Fig. 14.2). Normal red cells must undergo extreme deformation to leave the splenic red pulp cords and enter the sinusoids. Because of their spheroidal shape and reduced deformability, the hapless spherocytes are trapped in the cords, where they are easy prey for macrophages. The splenic environment also exacerbates the tendency of HS red cells to lose membrane along with K^+ ions and H_2O ; prolonged splenic exposure (erythrostasis), depletion of red cell glucose, and diminished red cell pH have all been suggested to contribute to these abnormalities. After splenectomy, the spherocytes persist, but the anemia is corrected.

MORPHOLOGY

The most specific morphologic finding is **spherocytosis**, apparent on smears as small, dark-staining (hyperchromic) red cells lacking the zone of central pallor (Fig. 14.3). Spherocytosis is distinctive but not pathognomonic, as spherocytes are also seen in other disorders associated with red cell membrane loss, such as in autoimmune hemolytic anemia. Other features are common to all hemolytic anemias, including marked variations in red cell size (anisocytosis). These include reticulocytosis, erythroid hyperplasia in the marrow, hemosiderosis, and mild jaundice. **Cholelithiasis** (pigment stones) occurs in 40% to 50% of affected adults. Moderate **splenomegaly** is characteristic (500 to 1000 g); in few other hemolytic anemias is the spleen enlarged as much or as consistently. Splenomegaly results from congestion of the red pulp cords and increased numbers of phagocytes.

Clinical Features

The diagnosis is based on family history, hematologic findings, and laboratory evidence. The combination of spherocytosis, reticulocytosis, and a negative direct Coombs test (the absence of antibody or complement on red cells) is highly suggestive of HS. Another clue is an

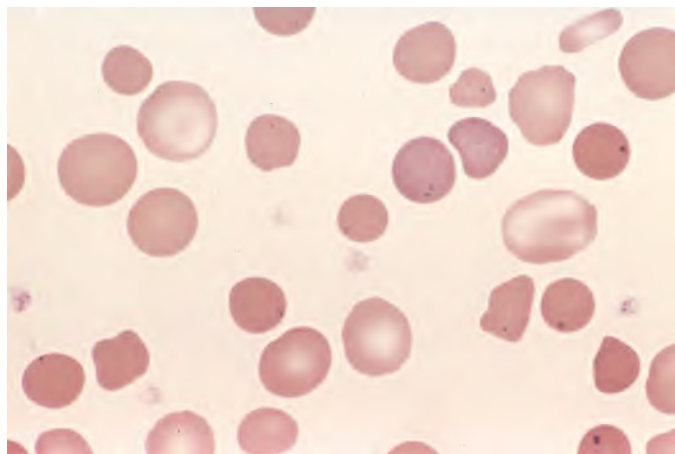


Fig. 14.3 Hereditary spherocytosis (peripheral smear). Note the variation in cell size (anisocytosis) and several dark spherocytes without central pallor. Howell-Jolly bodies (small, black nuclear remnants) are also seen in some of the red cells of this asplenic patient. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

increased mean cell hemoglobin concentration, due to dehydration of HS red cells caused by the loss of K^+ and H_2O . A useful confirmatory test involves incubation of red cells with a fluorescent dye, eosin-5'-maleimide, that selectively binds to band 3; staining is decreased in red cell disorders such as HS that lead to the loss of membrane.

The characteristic clinical features are anemia, splenomegaly, and jaundice. The severity varies greatly. In a small minority (mainly compound heterozygotes), HS presents at birth with marked jaundice and requires exchange transfusions. In 20% to 30% of patients, the decreased red cell survival is readily compensated for by increased erythropoiesis and the disease is so mild as to be virtually asymptomatic. In most, however, the compensatory changes are outpaced by hemolysis, producing chronic anemia of mild to moderate severity.

The generally stable clinical course is sometimes punctuated by *aplastic crises*, usually triggered by an acute parvovirus infection. Parvovirus infects and kills red cell progenitors, causing all red cell production to cease until an immune response clears the virus, generally in 1 to 2 weeks. Because of the reduced life span of HS red cells, cessation of erythropoiesis for even short periods leads to sudden worsening of the anemia. Transfusions may be necessary to support the patient during the acute phase of the infection. *Hemolytic crises* are produced by intercurrent events leading to increased splenic destruction of red cells (e.g., infectious mononucleosis and its attendant increase in spleen size); these are clinically less significant than aplastic crises. Gallstones, found in many patients, may also produce symptoms. Splenectomy treats the anemia and its complications but brings with it an increased risk of sepsis because the spleen acts as an important filter for blood-borne bacteria.

Hemolytic Disease Due to Red Cell Enzyme Defects: Glucose-6-Phosphate Dehydrogenase Deficiency

Abnormalities in the hexose monophosphate shunt or glutathione metabolism resulting from deficient or impaired enzyme function reduce the ability of red cells to protect themselves against oxidative injuries, leading to hemolysis. The most important of these enzyme derangements is hereditary deficiency of glucose-6-phosphate dehydrogenase (G6PD) activity. G6PD reduces nicotinamide adenine dinucleotide phosphate (NADP) to NADPH while oxidizing glucose-6-phosphate (Fig. 14.4). NADPH then provides reducing equivalents needed for conversion of oxidized glutathione to reduced glutathione, which protects against injury by participating as a cofactor in reactions that neutralize oxidants such as hydrogen peroxide (H_2O_2 ; see Fig. 14.4).

G6PD deficiency is a recessive X-linked trait, placing males at much higher risk for symptomatic disease. Several hundred G6PD genetic variants exist, but most clinically significant hemolytic anemia is associated with only two variants, designated G6PDA⁻ and G6PD Mediterranean. G6PDA⁻ is present in about 10% of Black Americans; G6PD Mediterranean, as the name implies, is prevalent in the Middle East. The high frequency of these variants in each population is believed to stem from a protective effect against *Plasmodium falciparum* malaria (discussed later). G6PD variants associated with hemolysis result in misfolding of the protein, making it more susceptible to proteolytic degradation. Compared with the most common normal variant, G6PD B, the half-life of G6PDA⁻ is moderately reduced, whereas that of G6PD Mediterranean is markedly decreased. Because mature red cells do not synthesize new proteins, as red cells age G6PDA⁻ and G6PD Mediterranean enzyme activities fall to levels that are inadequate to protect against oxidant stress. Thus, older red cells are much more prone to hemolysis than younger ones.

The episodic hemolysis that is characteristic of G6PD deficiency is caused by exposures that generate oxidant stress. The most common triggers are infections, in which oxygen-derived free radicals are

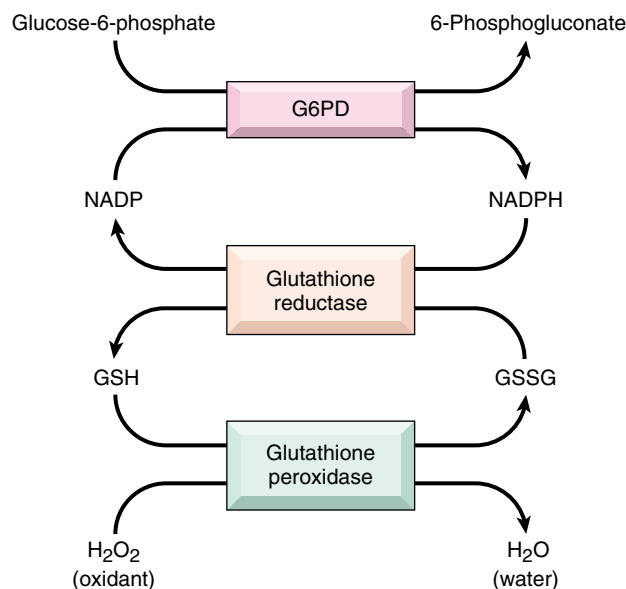


Fig. 14.4 Role of glucose-6-phosphate dehydrogenase (G6PD) in defense against oxidant injury. Detoxification of H_2O_2 , a potential oxidant, requires reduced glutathione (GSH), which is generated in a reaction that requires reduced nicotinamide adenine dinucleotide (NADPH). The synthesis of NADPH depends on the activity of G6PD. GSSG, Oxidized glutathione; NADP, nicotinamide adenine dinucleotide phosphate.

produced by activated leukocytes. Many infections can trigger hemolysis; viral hepatitis, pneumonia, and typhoid fever are among those most likely to do so. The other important initiators are drugs and certain foods. The drugs implicated are numerous, including antimalarials (e.g., primaquine and chloroquine), sulfonamides, nitrofurantoin, and others. Some drugs cause hemolysis only in individuals with the more severe Mediterranean variant. The most frequently cited food is the *fava bean*, which generates oxidants when metabolized. “Favism” is endemic in the Mediterranean, Middle East, and parts of Africa where consumption is prevalent. Uncommonly, G6PD deficiency presents as neonatal jaundice or a chronic low-grade hemolytic anemia in the absence of infection or known environmental triggers.

Oxidants cause both intravascular and extravascular hemolysis in G6PD-deficient individuals. Exposure of G6PD-deficient red cells to high levels of oxidants causes the cross-linking of reactive sulfhydryl groups on globin chains, which denature and form membrane-bound precipitates known as *Heinz bodies*. These are seen as dark inclusions within red cells stained with crystal violet (Fig. 14.5). Heinz bodies can damage the membrane sufficiently to cause intravascular hemolysis. Less severe membrane damage results in decreased red cell deformability. As inclusion-bearing red cells pass through the splenic cords, macrophages pluck out the Heinz bodies. As a result of membrane damage, some of these partially devoured cells retain an abnormal shape, appearing to have a bite taken out of them (see Fig. 14.5). Other less severely damaged cells become spherocytes due to loss of membrane surface area. Both bite cells and spherocytes are trapped in splenic cords and removed by phagocytes.

Acute intravascular hemolysis, marked by anemia, hemoglobinemia, and hemoglobinuria, usually begins 2 to 3 days following exposure of G6PD-deficient individuals to environmental triggers. Because only older red cells are at risk for lysis, the episode is self-limited, as hemolysis ceases when only younger G6PD-replete red cells remain (even if exposure to the trigger, e.g., an offending drug, continues). The recovery phase is heralded by reticulocytosis. Because hemolytic

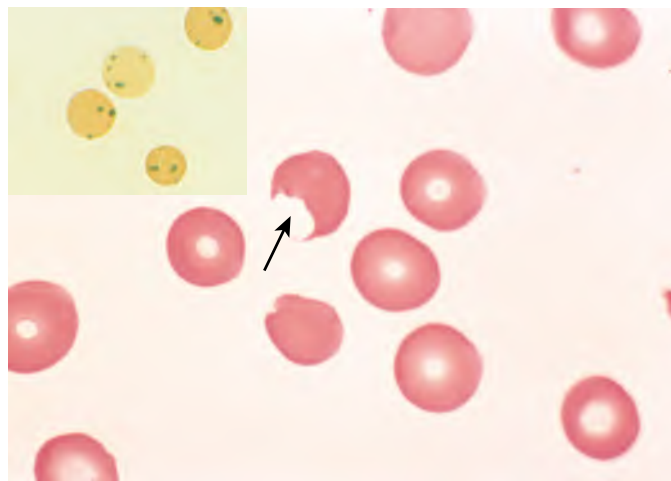


Fig. 14.5 Glucose-6-phosphate dehydrogenase deficiency: effects of oxidant drug exposure (peripheral blood smear). *Inset*, Red cells with precipitates of denatured globin (Heinz bodies) revealed by crystal violet staining. As the splenic macrophages pluck out these inclusions, “bite cells” (arrow) like the one in this smear are produced. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

episodes related to G6PD deficiency occur intermittently, features related to chronic hemolysis (e.g., splenomegaly, cholelithiasis) are absent.

Sickle Cell Disease

Sickle cell disease is a common hereditary hemoglobinopathy caused by a point mutation in β -globin that promotes the polymerization of deoxygenated hemoglobin, leading to red cell distortion, hemolytic anemia, microvascular obstruction, and ischemic tissue damage. Several hundred hemoglobinopathies caused by various mutations in globin genes are known, but only those associated with sickle cell disease are prevalent enough in the United States to merit discussion. Hemoglobin (Hb) is a tetrameric protein composed of two pairs of globin chains, each with its own heme group. Normal adult red cells contain mainly HbA ($\alpha_2\beta_2$), along with small amounts of HbA₂ ($\alpha_2\delta_2$) and fetal hemoglobin (HbF; $\alpha_2\gamma_2$). Sickle cell disease is caused by a missense mutation in the β -globin gene that leads to the replacement of a charged glutamate residue with a hydrophobic valine residue. The abnormal physiochemical properties of the resulting sickle hemoglobin (HbS) are responsible for the disease.

About 8% to 10% of African Americans in the United States are heterozygous for HbS, a largely asymptomatic condition known as *sickle cell trait*. The offspring of two heterozygotes has a 1 in 4 chance of being homozygous for the sickle mutation, a state that produces symptomatic sickle cell disease, which afflicts 70,000 to 100,000 individuals in the United States. In affected individuals, almost all the hemoglobin in the red cell is HbS ($\alpha_2\beta^s_2$).

The high prevalence of sickle cell trait in certain populations stems from its protective effects against falciparum malaria. Genetic studies have shown that the sickle hemoglobin mutation arose in West Africa and then spread to other populations, rising to high levels only in areas in which falciparum malaria is endemic, providing clear evidence of strong Darwinian selection. Parasite densities are lower in infected, heterozygous HbAS children than in infected, normal HbAA children, and HbAS children are significantly less likely to have severe disease or to die from malaria.

Although mechanistic details are lacking, two scenarios to explain these observations are favored:

- Metabolically active intracellular parasites consume oxygen and decrease intracellular pH, both of which promote sickling of HbAS red cells. These distorted, stiff cells may be cleared more rapidly by splenic and hepatic phagocytes, keeping parasite loads low.
- Sickling also impairs the formation of membrane knobs containing a protein made by the parasite called *PfEMP-1*. These membrane knobs are implicated in adhesion of infected red cells to endothelium, which is believed to have an important pathogenic role in the most severe form of the disease, cerebral malaria.

It has been suggested that G6PD deficiency and thalassemia also protect against malaria by increasing the clearance and decreasing the adherence of infected red cells, possibly by raising levels of oxidant stress and causing membrane damage in the parasite-bearing cells that leads to their rapid removal from the bloodstream.

Pathogenesis

The major pathologic manifestations—chronic hemolysis, microvascular occlusions, and tissue damage—all stem from the tendency of HbS molecules to stack into polymers when deoxygenated. Initially, this process converts the red cell cytosol from a freely flowing liquid into a viscous gel. With continued deoxygenation, HbS molecules assemble into long needlelike fibers within red cells, producing a distorted sickle or holly-leaf shape.

Several variables affect the rate and degree of sickling:

- **Interaction of HbS with the other types of hemoglobin.** In heterozygotes with sickle cell trait, about 40% of the hemoglobin is HbS and the rest is HbA, which interferes with HbS polymerization. As a result, red cells in heterozygous individuals only sickle if exposed to prolonged, relatively severe hypoxia. HbF inhibits the polymerization of HbS even more than HbA; hence, infants with sickle cell disease do not become symptomatic until they reach 5 or 6 months of age, when the level of HbF normally falls. However, in some individuals, HbF expression remains relatively high, a condition known as *hereditary persistence of fetal hemoglobin*; in these individuals, sickle cell disease is much less severe. Another variant hemoglobin, HbC, also is common in regions where HbS is found; overall, about 2% to 3% of Black Americans are HbC heterozygotes, and about 1 in 1250 are compound HbS/HbC heterozygotes. In HbSC red cells, the percentage of HbS is 50%, as compared with only 40% in HbAS cells. Moreover, with aging HbSC cells tend to lose salt and water and become dehydrated, an effect that increases the intracellular concentration of HbS. These factors increase the tendency for HbS to polymerize, and as a result compound HbSC heterozygotes have a symptomatic sickling disorder termed *HbSC disease* that is somewhat milder than sickle cell disease.
- **Mean cell hemoglobin concentration (MCHC).** Higher HbS concentrations increase the probability that aggregation and polymerization will occur during any given period of deoxygenation. Thus, intracellular dehydration, which increases the MCHC, facilitates sickling. Conversely, conditions that decrease the MCHC reduce disease severity. This occurs when an individual who is homozygous for HbS also has coexistent α -thalassemia, which reduces Hb synthesis and leads to milder disease.
- **Intracellular pH.** A decrease in pH reduces the oxygen affinity of hemoglobin, thereby increasing the fraction of deoxygenated HbS at any given oxygen tension and augmenting the tendency for sickling.
- **Transit time of red cells through microvascular beds.** As will be discussed, much of the pathology of sickle cell disease is related to

vascular occlusion caused by sickling within microvascular beds. Transit times in most normal microvascular beds are too short for significant aggregation of deoxygenated HbS to occur, and as a result sickling is confined to microvascular beds with slow transit times. Blood flow is sluggish in the normal spleen and bone marrow, which are predominantly affected in sickle cell disease. The movement of blood through inflamed tissues is slowed because of vasodilation and the transudation of fluid through leaky vessels.

Sickling causes cumulative damage to red cells through several mechanisms. As HbS polymers grow, they herniate through the membrane skeleton and project from the cell ensheathed only by the lipid bilayer. This severe derangement in membrane structure causes an influx of Ca^{2+} ions; the calcium ions induce the cross-linking of membrane proteins and activate an ion channel that leads to the efflux of K^{+} and H_2O . As a result, with repeated sickling episodes, red cells become dehydrated, dense, and rigid (Fig. 14.6). Eventually, the most severely damaged cells are converted to nondeformable irreversibly sickled cells that retain a sickle shape even when fully oxygenated. The severity of the hemolysis correlates with the percentage of irreversibly sickled cells, which are rapidly sequestered and removed by mononuclear phagocytes (extravascular hemolysis). Sickled red cells are also mechanically fragile, leading to some intravascular hemolysis as well.

The pathogenesis of the microvascular occlusions, which are responsible for the most serious clinical features, is less certain. Microvascular occlusions are not related to the number of irreversibly sickled cells, but instead may be dependent on more subtle red cell membrane damage and local factors, such as inflammation or vasoconstriction, that tend to slow or arrest the movement of red cells through microvascular beds (see Fig. 14.6). As mentioned earlier, sickle red cells express increased amounts of adhesion molecules and are sticky.

Mediators released from granulocytes during inflammatory reactions up-regulate the expression of adhesion molecules on endothelial cells (Chapter 3) and further enhance the tendency for sickle red cells to arrest during transit through the microvasculature. The stagnation of red cells within inflamed vascular beds results in extended exposure to low oxygen tension, sickling, and vascular obstruction. Once started, it is easy to envision how a vicious cycle of sickling, obstruction, hypoxia, and more sickling ensues. Depletion of nitric oxide (NO) also may play a part in the vascular occlusions. Free hemoglobin released from lysed sickle red cells can bind and inactivate NO, a potent vasodilator and inhibitor of platelet aggregation. This in turn may lead to increased vascular tone (narrowing of vessels) and enhanced platelet aggregation, both of which may contribute to red cell stasis, sickling, and (in some instances) thrombosis.

MORPHOLOGY

In sickle cell anemia, the peripheral blood demonstrates variable numbers of **irreversibly sickled cells**, reticulocytosis, and target cells, which result from red cell dehydration (Fig. 14.7). **Howell-Jolly bodies** (small nuclear remnants) are also present in red cells due to asplenia (see later). The bone marrow is hyperplastic owing to compensatory erythroid hyperplasia. Marked expansion of the marrow leads to bone resorption and secondary new bone formation, producing prominent cheekbones and changes in the skull that resemble a “crewcut” on radiographic studies. Extramedullary hematopoiesis may also appear. The increased breakdown of hemoglobin may cause hyperbilirubinemia and formation of pigment gallstones.

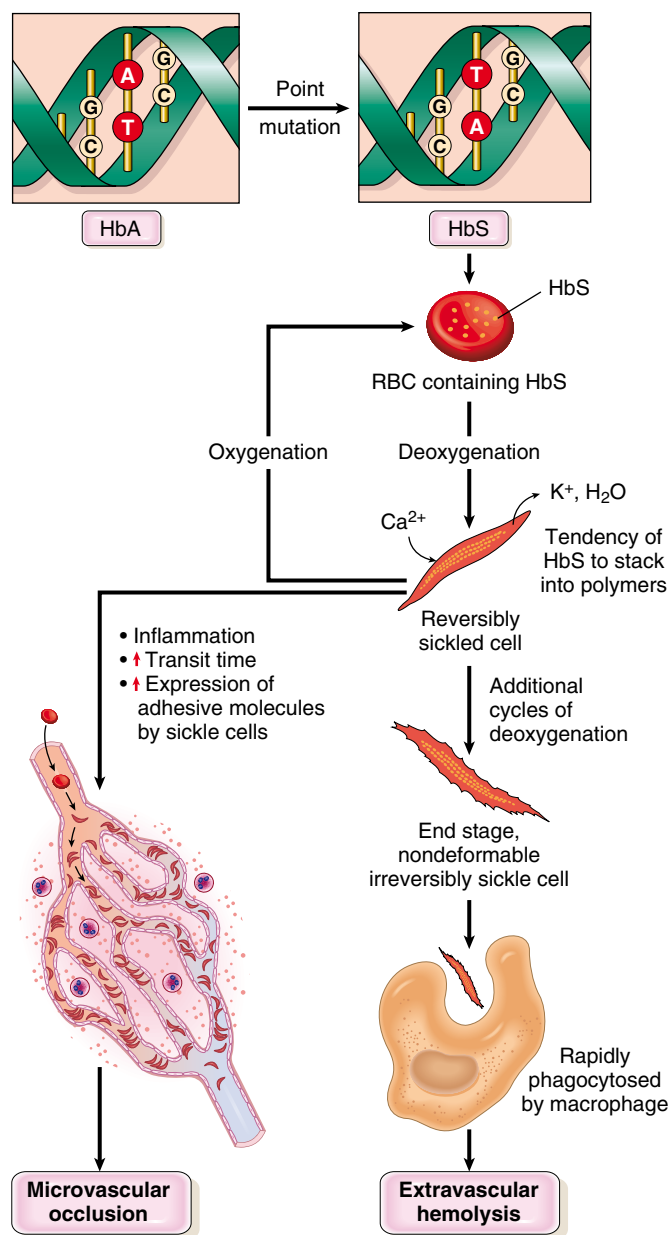


Fig. 14.6 Pathophysiology of sickle cell disease. *HbA*, Hemoglobin A; *HbS*, hemoglobin S; *RBC*, red blood cell.

In early childhood, the spleen is enlarged (up to 500 g) by red pulp congestion caused by the trapping of sickled red cells in the cords and sinuses (Fig. 14.8). With time, however, chronic erythroblastosis leads to splenic infarction, fibrosis, and progressive shrinkage, so that by adolescence or early adulthood only a small nubbin of fibrotic splenic tissue is left, a process called **autosplenectomy** (Fig. 14.9). Infarctions caused by vascular occlusions may occur in many other tissues as well, including the bones, brain, kidney, liver, retina, and pulmonary vessels, the latter sometimes producing cor pulmonale. In adult patients, vascular stagnation in subcutaneous tissues often leads to leg ulcers; this complication is rare in children.

Clinical Features

Sickle cell disease causes a moderately severe hemolytic anemia (hematocrit 18% to 30%) associated with reticulocytosis, hyperbilirubinemia,

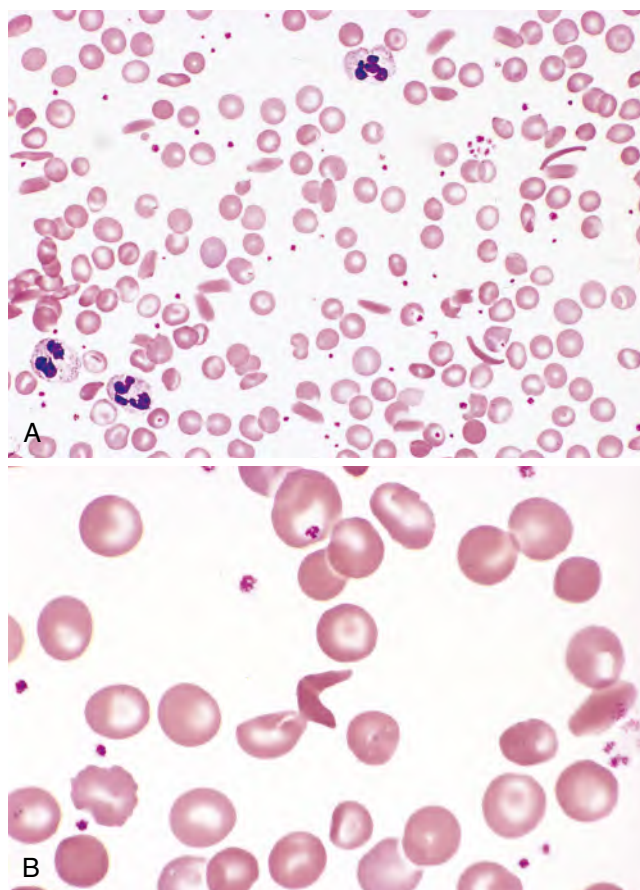


Fig. 14.7 Sickle cell disease (peripheral blood smear). (A) Low magnification shows irreversibly sickled cells as well as target cells and red cell anisocytosis and poikilocytosis. (B) Higher magnification shows an irreversibly sickled cell in the center. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

and the presence of irreversibly sickled cells. Its course is punctuated by a variety of “crises.” *Vaso-occlusive crises*, also called *pain crises*, are episodes of hypoxic injury and infarction that cause severe pain in the affected region. Although infection, dehydration, and acidosis (all favoring sickling) may act as triggers, in most instances no predisposing cause is identified. The most involved sites are the bones, lungs, liver, brain, spleen, and penis. In children, painful bone crises are extremely common and often difficult to distinguish from acute osteomyelitis. These frequently manifest as the *hand-foot syndrome* or dactylitis of the bones of the hands and feet. *Acute chest syndrome* is a particularly dangerous type of vaso-occlusive crisis involving the lungs that typically presents with fever, cough, chest pain, and pulmonary infiltrates. Pulmonary inflammation (such as induced by infection) may cause blood flow to become sluggish and “spleenlike,” leading to sickling and vaso-occlusion. This compromises pulmonary function, creating a potentially fatal cycle of worsening pulmonary and systemic hypoxemia, sickling, and vaso-occlusion. *Priapism* affects up to 45% of males after puberty and may lead to hypoxic damage and erectile dysfunction. Other disorders related to vascular obstruction, particularly stroke and retinopathy leading to loss of visual acuity and even blindness, can take a devastating toll. Factors proposed to contribute to stroke include the adhesion of sickle red cells to arterial vascular endothelium and vasoconstriction caused by the depletion of NO by free hemoglobin.

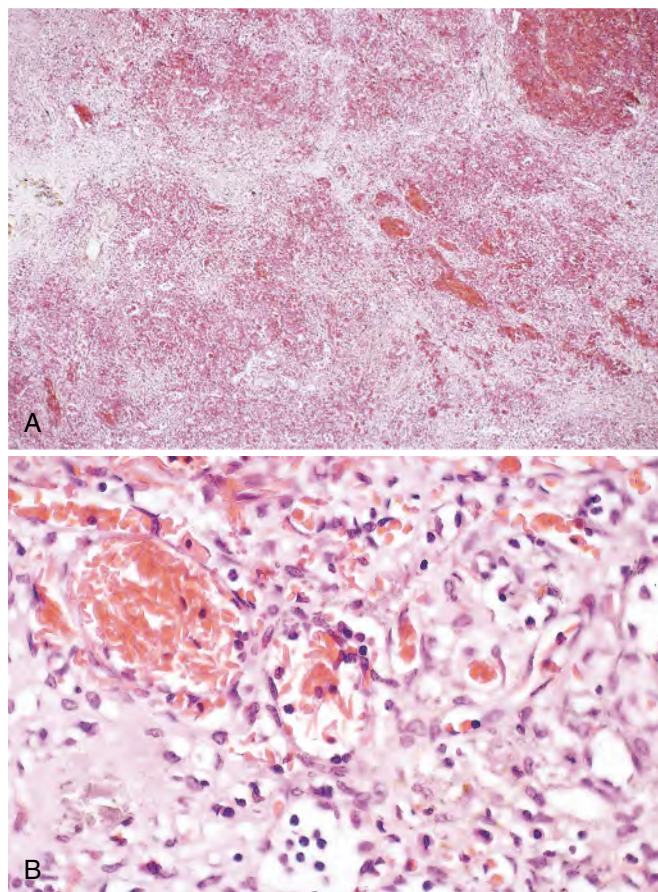


Fig. 14.8 (A) Spleen in sickle cell disease (low power). Red pulp cords and sinusoids are markedly congested; between the congested areas, pale areas of fibrosis resulting from ischemic damage are evident. (B) Under high power, splenic sinusoids are dilated and filled with sickled red cells. (Courtesy Dr. Darren Wirthwein, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)



Fig. 14.9 “Autoinfarcted” splenic remnant in sickle cell disease. (Courtesy Drs. Dennis Burns and Darren Wirthwein, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Although occlusive crises are the most common cause of patient morbidity and mortality, several other acute events complicate the course. *Sequestration crises* occur in children with intact spleens. Massive entrapment of sickled red cells leads to rapid splenic enlargement, hypovolemia, and sometimes shock. Both sequestration crises and the

acute chest syndrome may be fatal and sometimes require prompt treatment with exchange transfusions. *Aplastic crises* stem from the infection of red cell progenitors by parvovirus B19, which causes a transient cessation of erythropoiesis and a sudden worsening of the anemia.

In addition to these dramatic crises, chronic tissue hypoxia takes a more subtle but nevertheless important toll. Chronic hypoxia impairs growth and development and damages the spleen, heart, kidneys, and lungs. Sickling provoked by hypertonicity in the renal medulla causes damage that eventually leads to *hyposthenuria* (the inability to concentrate urine), which increases the propensity for dehydration and its attendant risks.

Increased susceptibility to infection with encapsulated organisms is another threat. This is due in large part to altered splenic function, which is severely impaired in children by congestion and poor blood flow, and completely absent in adults because of splenic infarction. Defects in the alternative complement pathway of uncertain etiology also impair the opsonization of bacteria. *Pneumococcus pneumoniae* and *Haemophilus influenzae* septicemia and meningococcal meningitis are common, particularly in children, but their incidence can be reduced by vaccination and prophylactic antibiotics.

It must be emphasized that there is great variation in the clinical manifestations of sickle cell disease. Some individuals experience repeated vaso-occlusive crises, whereas others have only mild symptoms. The basis for this wide range in disease expression is not understood; both modifying genes and environmental factors are suspected.

The diagnosis is suggested by the clinical findings and the presence of irreversibly sickled red cells and is confirmed by various tests for sickle hemoglobin. Prenatal diagnosis is possible by analysis of fetal DNA obtained by amniocentesis or chorionic biopsy. Newborn screening for sickle hemoglobin is now routinely performed in all 50 states, typically using samples obtained by heel stick at birth.

The outlook for patients with sickle cell disease has improved considerably over the past several decades. About 90% of patients survive to 20 years of age, and close to 50% survive beyond the fifth decade. The mainstay of treatment is an inhibitor of DNA synthesis, hydroxyurea, which has several beneficial effects. These include (1) an increase in red cell HbF levels, which occurs by unknown mechanisms; and (2) an antiinflammatory effect, which stems from an inhibition of leukocyte production. These activities (and possibly others) are believed to act in concert to decrease crises related to vascular occlusions in both children and adults. When added to hydroxyurea, L-glutamine appears to decrease pain crises; the mechanism is uncertain but may involve changes in metabolism that decrease red cell oxidant stress. Hematopoietic stem cell transplantation offers a chance at cure in those with disease that is not well controlled with other treatments. An exciting new approach involving the use of gene editing (CRISPR technology) to increase the production of fetal hemoglobin and prevent sickling has recently been approved for clinical use. Although curative, this approach is very costly, and other more affordable approaches will need to be developed to treat patients in low-resource countries.

Thalassemia

Thalassemia is a genetically heterogeneous disorder caused by germline mutations that decrease the synthesis of either α -globin or β -globin, leading to anemia, tissue hypoxia, and red cell hemolysis related to the imbalance in globin chain synthesis. The two α chains in HbA are encoded by an identical pair of α -globin genes on chromosome 16, and the two β chains are encoded by a single β -globin gene on chromosome 11. β -thalassemia is caused by deficient synthesis of β chains, whereas α -thalassemia is caused by deficient synthesis of α chains. The hematologic consequences of diminished synthesis

of one globin chain stem not only from hemoglobin deficiency but also from a relative excess of the other globin chain, particularly in β -thalassemia (described later).

Thalassemia is endemic in the Mediterranean basin (indeed, *thalassa* means “sea” in Greek) as well as the Middle East, tropical Africa, the Indian subcontinent, and Asia, and in aggregate is among the most common inherited disorders of humans. As with sickle cell disease and other common inherited red cell disorders, its prevalence seems to be explained by the protection it affords heterozygous carriers against malaria. Although we discuss thalassemia with other inherited forms of anemia associated with hemolysis, it is important to recognize that the defects in globin synthesis that underlie these disorders cause anemia through two mechanisms, decreased red cell production and decreased red cell lifespan.

β -Thalassemia. β -thalassemia is caused by mutations that diminish the synthesis of β -globin chains. Its clinical severity varies widely due to heterogeneity in the causative mutations. We will begin our discussion with the molecular lesions in β -thalassemia and then relate the clinical variants to specific underlying molecular defects.

Molecular Pathogenesis

The causative mutations fall into two categories: (1) β^0 mutations, associated with absent β -globin synthesis, and (2) β^+ mutations, characterized by reduced (but detectable) β -globin synthesis. Sequencing of β -thalassemia genes has revealed more than 100 different causative mutations, mostly consisting of point mutations, which fall into three major classes:

- **Splicing mutations.** These are the most common cause of β^+ -thalassemia. Some of these mutations destroy normal RNA splice junctions and completely prevent the production of normal β -globin messenger RNA (mRNA), resulting in β^0 -thalassemia. Others create an “ectopic” splice site within an intron. Because the flanking normal splice site remains, both normal and abnormal splicing occurs and some normal β -globin mRNA is made, resulting in β^+ -thalassemia.
- **Promoter region mutations.** These mutations reduce transcription by 75% to 80%. Because some normal β -globin is synthesized, these mutations are associated with β^+ -thalassemia.
- **Chain terminator mutations.** These are the most common cause of β^0 -thalassemia. They consist of either nonsense mutations that introduce a premature stop codon or small insertions or deletions that shift the mRNA reading frames (frameshift mutations; Chapter 5). Both block translation and prevent the synthesis of any functional β -globin.

Impaired β -globin synthesis results in anemia by two mechanisms (Fig. 14.10). The deficit in HbA synthesis produces “underhemoglobinized” hypochromic, microcytic red cells with subnormal oxygen transport capacity. Even more important is the diminished survival of red cells and their precursors, which results from the imbalance in α - and β -globin synthesis. Unpaired α chains precipitate within red cell precursors, forming insoluble inclusions. These inclusions cause a variety of untoward effects, but membrane damage is the proximal cause of most red cell pathology. Many red cell precursors succumb to membrane damage and undergo apoptosis. In severe β -thalassemia, it is estimated that 70% to 85% of red cell precursors suffer this fate, which leads to *ineffective erythropoiesis*. Those red cells that are released from the marrow also contain inclusions and have membrane damage, leaving them prone to splenic sequestration and extravascular hemolysis.

In severe β -thalassemia, ineffective erythropoiesis creates several additional problems. Erythropoietic drive in the setting of severe

uncompensated anemia leads to massive erythroid hyperplasia in the marrow and extensive extramedullary hematopoiesis. The expanding mass of red cell precursors erodes the bony cortex, impairs bone growth, and produces skeletal abnormalities (described later). Extramedullary hematopoiesis involves the liver, spleen, and lymph nodes, and in extreme cases produces extraosseous masses in the thorax, abdomen, and pelvis. The metabolically active erythroid progenitors steal nutrients from other tissues that are already oxygen-starved, causing severe cachexia in untreated patients.

Another serious complication of ineffective erythropoiesis is excessive absorption of dietary iron. Erythroid precursors secrete a hormone called *erythroferrone* that inhibits production of *hepcidin*, a key negative regulator of iron uptake in the gut (described later). In thalassemia, the marked expansion of erythroid precursors leads to increased absorption of iron from the gut (Fig. 14.11), and this, together with repeated blood transfusions, inevitably leads to severe iron accumulation (*secondary hemochromatosis*) unless preventive steps are taken. Injury to parenchymal organs, particularly the heart and liver, often follows (Chapter 18).

Clinical Syndromes

The relationships of clinical phenotypes to underlying genotypes are summarized in Table 14.3. Clinical classification of β -thalassemia is based on the severity of the anemia, which in turn depends on the genetic defect (β^+ or β^0) and the gene dosage (homozygous or heterozygous). In general, individuals with two β -thalassemia alleles (β^+/β^+ , β^+/β^0 , or β^0/β^0) have a severe, transfusion-dependent anemia called β -thalassemia major. Heterozygotes with one β -thalassemia gene and one normal gene (β^+/β or β^0/β) usually have a mild asymptomatic microcytic anemia. This condition is referred to as β -thalassemia minor or β -thalassemia trait. A third genetically heterogeneous variant of moderate severity is called β -thalassemia intermedia. This category includes milder variants of β^+/β^+ or β^+/β^0 -thalassemia and unusual forms of heterozygous β -thalassemia. Some patients with β -thalassemia intermedia have two defective β -globin genes and an α -thalassemia gene defect, which improves the effectiveness of erythropoiesis and red cell survival by lessening the imbalance in α - and β -chain synthesis. In other rare but informative cases, affected individuals have a single β -globin defect and one or two extra copies of normal α -globin genes (stemming from a gene duplication event), which worsens the chain imbalance. These unusual forms of the disease emphasize the cardinal role of unpaired α -globin chains in the pathology. The clinical and morphologic features of β -thalassemia intermedia are not described separately but can be surmised from the following discussions of β -thalassemia major and β -thalassemia minor.

β -Thalassemia Major. β -thalassemia major is most common in Mediterranean countries, parts of Africa, and Southeast Asia. In the United States, the incidence is highest in immigrants from these areas. The anemia manifests 6 to 9 months after birth as hemoglobin synthesis switches from HbF to HbA. In untransfused patients, hemoglobin levels are 3 to 6 g/dL. The red cells may completely lack HbA (β^0/β^0 genotype) or contain small amounts (β^+/β^+ or β^0/β^+ genotypes). The major red cell hemoglobin is HbF, which is markedly elevated.

MORPHOLOGY

Blood smears show severe red cell abnormalities, including marked variation in size (**anisocytosis**) and shape (**poikilocytosis**), **microcytosis**, and **hypochromia**. Target cells (so called because hemoglobin collects in the center of the cell), basophilic stippling, and fragmented red cells are also common. Inclusions of aggregated α chains are efficiently removed by the spleen and

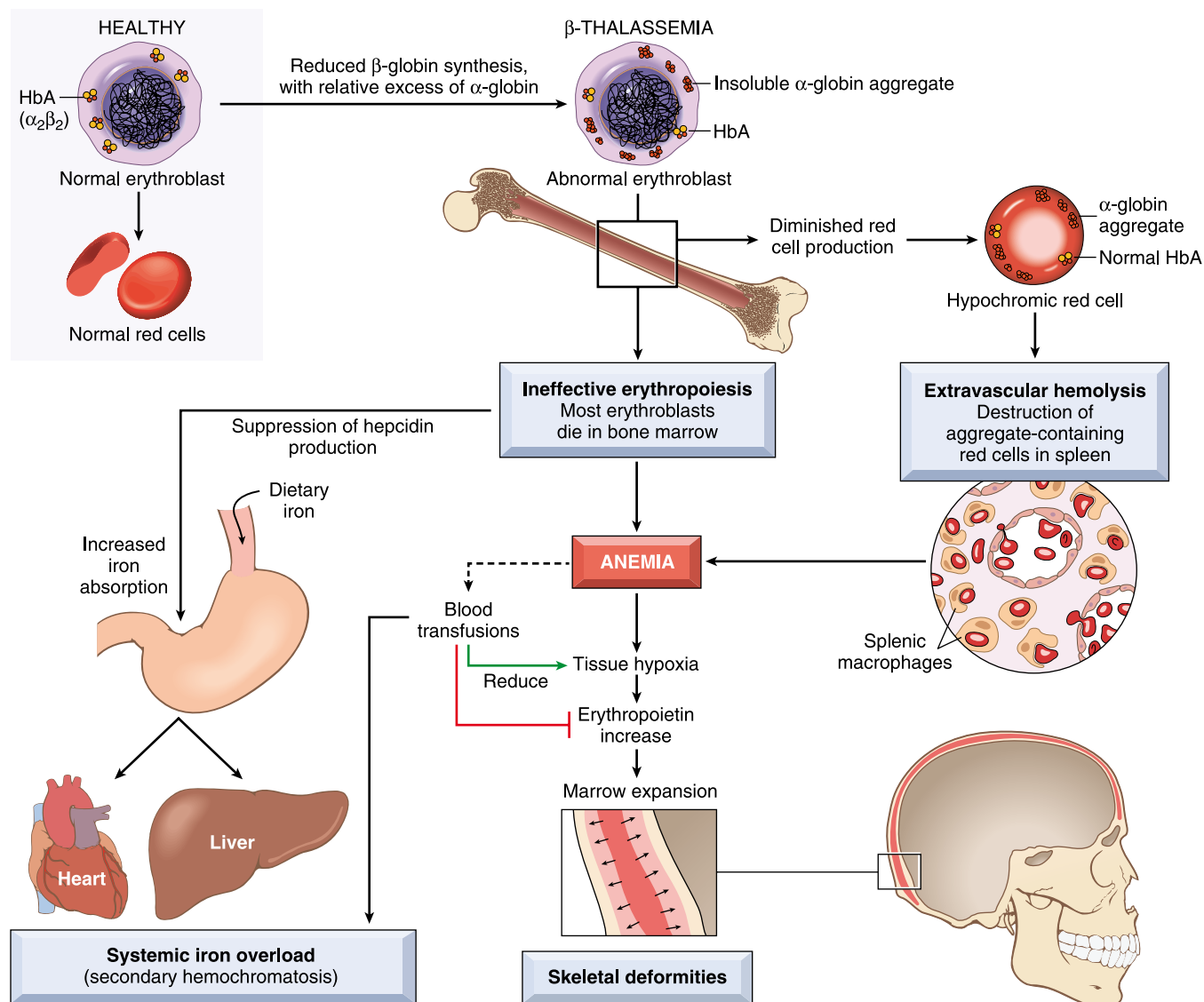


Fig. 14.10 Pathogenesis of β-thalassemia major. Note that the aggregates of unpaired α-globin chains, a hallmark of the disease, are not visible in routinely stained blood smears. Blood transfusions are a double-edged sword, diminishing the anemia and its attendant complications, but also adding to the systemic iron overload. HbA, Hemoglobin A.

not easily seen. The reticulocyte count is elevated but is lower than expected for the severity of anemia because of ineffective erythropoiesis. Variable numbers of poorly hemoglobinized nucleated red cell precursors (normoblasts) are seen in the peripheral blood due to “stress” erythropoiesis and abnormal release of red cell precursors from sites of extramedullary hematopoiesis.

Other major alterations involve the bone marrow and spleen. In untransfused patients, there is a striking expansion of hematopoietically active marrow. In facial bones and the skull, the burgeoning marrow erodes cortical bone and induces new bone formation, giving rise to a “crewcut” appearance on radiographic studies (Fig. 14.12). Both phagocyte hyperplasia and extramedullary hematopoiesis contribute to enlargement of the spleen, which can weigh as much as 1500 g. The liver and the lymph nodes may also be enlarged by extramedullary hematopoiesis.

Hemosiderosis and **secondary hemochromatosis**, the two manifestations of iron overload (Chapter 18), inevitably occur unless chelation therapy is given. The deposited iron often damages organs, most notably the heart, liver, and pancreas.

Clinical Features

The clinical course of β-thalassemia major is brief unless blood transfusions are given. Untreated children experience growth retardation and die at an early age from the effects of anemia. In those who survive long enough, the cheekbones and other bony prominences are enlarged and distorted. Hepatosplenomegaly due to extramedullary hematopoiesis is usually present. Although blood transfusions improve the anemia and suppress complications related to excessive erythropoiesis, they lead to complications of their own. Cardiac disease resulting from progressive iron overload and secondary hemochromatosis (Chapter 18) is an important cause of death, particularly in heavily transfused patients, who must be treated with iron chelators to prevent this complication. With transfusions and iron chelation, survival into the third decade is possible, but the overall outlook remains guarded. Hematopoietic stem cell transplantation can be curative and is being used increasingly. Gene therapy involving the use of a lentivirus to introduce a functional β-globin gene into patient stem cells is now approved. As with CRISPR therapy,

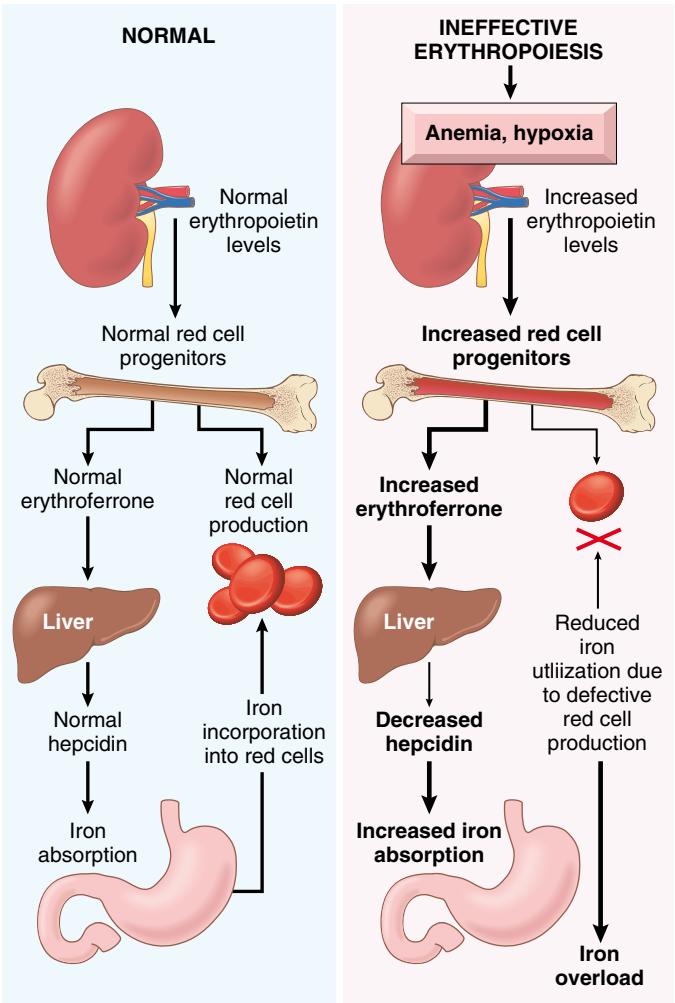


Fig. 14.11 Mechanism of iron overload due to ineffective hematopoiesis. In the setting of ineffective erythropoiesis, such as in those with severe thalassemia, increased release of erythroferrone from the expanded mass of erythroid progenitors suppresses hepcidin production, leading to increased iron uptake from the gut.

however, the cost of this treatment is very high, precluding use in low-resource parts of the world. Prenatal diagnosis is possible by molecular analysis of DNA.

β -Thalassemia Minor. β -thalassemia minor is much more common than β -thalassemia major and understandably affects the same ethnic groups. Most patients are heterozygous carriers of a β^+ or β^0 allele. These patients are usually asymptomatic. Anemia, if present, is mild. The peripheral blood smear typically shows hypochromia, microcytosis, basophilic stippling, and target cells. Mild erythroid hyperplasia is seen in the bone marrow. Hemoglobin electrophoresis usually reveals an increase in HbA₂ ($\alpha_2\delta_2$) to 4% to 8% of the total hemoglobin (normal, $2.5\% \pm 0.3\%$), reflecting an elevated ratio of δ -chain to β -chain synthesis. HbF levels are generally normal or occasionally slightly increased.

Recognition of β -thalassemia trait is important because it may be mistaken for iron deficiency and has implications for genetic counseling. Iron deficiency (the most common cause of microcytic anemia) can usually be excluded by measurement of serum iron, total iron-binding capacity, and serum ferritin (described under Iron Deficiency Anemia later in this chapter). The increase in HbA₂ is diagnostically useful, particularly in individuals (such as females of childbearing age) who are at high risk of iron deficiency.

α -Thalassemia. α -Thalassemia is caused by inherited deletions that result in reduced or absent synthesis of α -globin chains. Normal individuals have four α -globin genes, and the severity of α -thalassemia depends on how many α -globin genes are affected. As in β -thalassemia, the anemia stems from both inadequate hemoglobin synthesis and the presence of excess, unpaired β , γ , and δ globin chains, which vary in type at different ages. In newborns with α -thalassemia, excess unpaired γ -globin chains form γ_4 tetramers known as *hemoglobin Barts*, whereas in older children and adults, excess β -globin chains form β_4 tetramers known as *HbH*. Because free β and γ chains are more soluble than free α chains and form relatively stable homotetramers, hemolysis and ineffective erythropoiesis are less severe than in β -thalassemia.

Clinical Syndromes

The clinical syndromes are determined and classified by the number of α -globin genes that are deleted. Each of the four α -globin genes normally contributes 25% of the total α -globin chains. α -Thalassemia syndromes stem from combinations of deletions that remove one to

TABLE 14.3 Clinical and Genetic Classification of Thalassemia

Clinical Syndrome	Genotype	Clinical Features	Molecular Genetics
β-Thalassemia			
β-Thalassemia major	β°/β°, β+/β+, β°/β+	Severe; requires blood transfusions	Mainly point mutations that lead to defects in the transcription, splicing, or translation of β-globin mRNA
β-Thalassemia intermedia	Variable (β°/β+, β+/β+, β°/β, β+/β)	Severe but does not require regular blood transfusions	
β-Thalassemia minor	β°/β, β+/β	Asymptomatic with mild or absent anemia; red cell abnormalities seen	
α-Thalassemia			
Silent carrier	−/α α/α	Asymptomatic; no red cell abnormality	Mainly gene deletions
α-Thalassemia trait	−/− α/α (Asian) −/α −/α (Black African, Asian)	Asymptomatic, like β-thalassemia minor	
HbH disease	−/− −/α	Severe; resembles β-thalassemia intermedia	
Hydrops fetalis	−/− −/− β, wild type; β+, β+ allele; β°, β° allele	Lethal in utero without transfusions	

HbH disease, Hemoglobin H disease.



Fig. 14.12 β -Thalassemia major. X-ray film of the skull showing new bone formation on the outer table, producing perpendicular radiations resembling a crewcut. (Courtesy Dr. Jack Reynolds, Department of Radiology, University of Texas Southwestern Medical School, Dallas.)

four α -globin genes. Not surprisingly, the severity of the clinical syndrome is proportional to the number of α -globin genes that are deleted. The different types of α -thalassemia and their salient clinical features are listed in Table 14.3.

Silent carrier state. Silent carrier state is associated with the deletion of a single α -globin gene, which causes a barely detectable reduction in α -globin chain synthesis. These individuals are completely asymptomatic but have slight microcytosis.

α -Thalassemia Trait. α -Thalassemia trait is caused by the deletion of two α -globin genes from a single chromosome (α/α $-/-$) or the deletion of one α -globin gene from each of the two chromosomes ($\alpha/-$ $\alpha/-$) (see Table 14.3). The former genotype is more common in Asian populations, the latter in people from regions of Africa. Both genotypes produce similar deficiencies of α -globin, but they have different implications for the children of affected individuals, who are at risk of clinically significant α -thalassemia (HbH disease or α -thalassemia major) only when at least one parent has the $-/-$ haplotype. As a result, symptomatic α -thalassemia is relatively common in Asian populations and rare in African populations. The clinical picture in α -thalassemia trait is identical to that described for β -thalassemia minor, that is, small red cells (microcytosis), minimal or no anemia, and no abnormal physical signs. HbA₂ levels are normal or low.

Hemoglobin H Disease. Hemoglobin H (HbH) disease is caused by deletion of three α -globin genes. It is most common in Asian populations. With only one normal α -globin gene, the synthesis of α chains is markedly reduced, and tetramers of β -globin, called HbH, form. HbH has an extremely high affinity for oxygen and delivers little oxygen to tissues, leading to tissue hypoxia disproportionate to the level of hemoglobin. Additionally, HbH is prone to oxidation, which causes it to precipitate and form intracellular inclusions that promote red cell sequestration and phagocytosis in the spleen. The result is a moderately severe anemia resembling β -thalassemia intermedia.

Alpha Thalassemia Major. Alpha thalassemia major (previously called hydrops fetalis), the most severe form of α -thalassemia, is caused by deletion of all four α -globin genes. In the fetus, excess γ -globin chains form tetramers (hemoglobin Barts) with such high affinity for oxygen that they deliver little to tissues. Survival in early development is due to the expression of ζ chains, an embryonic globin

that pairs with γ chains to form a functional $\zeta_2\gamma_2$ Hb tetramer. Signs of fetal distress usually become evident by the third trimester of pregnancy. In the past, severe tissue anoxia led to death in utero or shortly after birth; with intrauterine transfusion many affected infants are now saved. The fetus shows severe pallor, generalized edema, and massive hepatosplenomegaly like that seen in hemolytic disease of the newborn (Chapter 10). There is a lifelong dependence on blood transfusions for survival, with the associated risk of iron overload. Hematopoietic stem cell transplantation can be curative.

Paroxysmal Nocturnal Hemoglobinuria

Paroxysmal nocturnal hemoglobinuria (PNH) results from acquired mutations in the phosphatidylinositol glycan complementation group A gene (*PIGA*), an enzyme that is essential for the synthesis of certain membrane-associated complement regulatory proteins. PNH has an incidence of 2 to 5 per million in the United States. Despite its rarity, it has fascinated hematologists because it is the only hemolytic anemia caused by an acquired genetic defect. Recall that proteins are anchored into the lipid bilayer in two ways. Most have a hydrophobic region that spans the cell membrane; these are called transmembrane or integral membrane proteins. The others are attached to the cell membrane through a covalent linkage to a specialized phospholipid called glycosylphosphatidylinositol (GPI). In PNH, these GPI-linked proteins are deficient because of somatic mutations that inactivate *PIGA*. *PIGA* is X-linked and subject to lyonization (random inactivation of one X chromosome in cells of females; Chapter 5). As a result, a single acquired mutation in the active *PIGA* gene of any given cell is sufficient to produce a deficiency state. Because the causative mutations occur in a hematopoietic stem cell, all its clonal progeny (red cells, white cells, and platelets) are deficient in GPI-linked proteins. Typically, only a subset of stem cells acquires the mutation, and the mutant clone coexists with the progeny of normal stem cells that are not *PIGA* deficient.

Remarkably, most normal individuals harbor small numbers of bone marrow cells with *PIGA* mutations identical to those that cause PNH. It is hypothesized that these cells increase in numbers (thus producing clinically evident PNH) only in rare instances where they have a selective advantage, such as in the setting of autoimmune reactions against GPI-linked antigens. Such a scenario might explain the frequent association of PNH and aplastic anemia, a marrow failure syndrome (discussed later) that has an autoimmune basis in many individuals.

PNH blood cells are deficient in three GPI-linked proteins that regulate complement activity: (1) decay-accelerating factor, or CD55; (2) membrane inhibitor of reactive lysis, or CD59; and (3) C8-binding protein. Of these factors, the most important is CD59, a potent inhibitor of C3 convertase that prevents the spontaneous activation of the alternative complement pathway.

Red cells deficient in GPI-linked factors are abnormally susceptible to lysis or injury by complement. This manifests as intravascular hemolysis, which is caused by the C5b-9 membrane attack complex. The hemolysis is paroxysmal and nocturnal in only 25% of cases; chronic hemolysis without dramatic hemoglobinuria is more typical. The tendency for red cells to lyse at night is explained by a slight decrease in blood pH during sleep, which increases the activity of complement. The anemia is variable but usually mild to moderate in severity. The loss of heme iron in the urine (hemosiderinuria) eventually leads to iron deficiency, which can exacerbate the anemia if untreated.

Thrombosis is the leading cause of disease-related death in individuals with PNH. About 40% of patients have venous thrombosis, often involving the hepatic, portal, or cerebral veins. How complement activation leads to thrombosis in patients with PNH is not clear; the absorption of NO by free hemoglobin (discussed under Sickle Cell

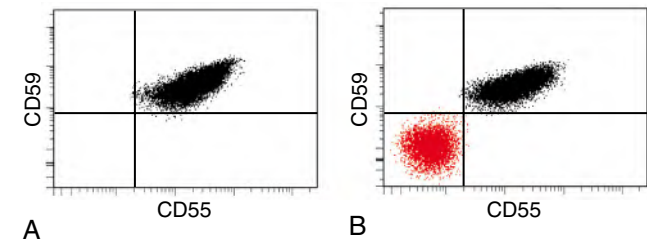


Fig. 14.13 Paroxysmal nocturnal hemoglobinuria (PNH). (A) Flow cytogram of blood from a normal individual shows that the red cells express two phosphatidylinositol glycan (PIG)-linked membrane proteins, CD55 and CD59, on their surfaces. (B) Flow cytogram of blood from a patient with PNH shows a population of red cells that is deficient in both CD55 and CD59. As is typical of PNH, a second population of CD55+/CD59+ red cells that is derived from residual normal hematopoietic stem cells is also present. (Courtesy Dr. Scott Rodig, Department of Pathology, Brigham and Women’s Hospital, Boston.)

Disease earlier) may be one contributing factor, and a role for endothelial damage caused by the C5-9 membrane attack complex is also suspected.

About 5% of patients with PNH eventually develop a myeloid neoplasm, typically either acute myeloid leukemia or a myelodysplastic neoplasm, indicating that PNH may arise in the context of genetic damage to hematopoietic stem cells.

PNH is diagnosed by flow cytometry, which provides a sensitive means for detecting red cells that are deficient in GPI-linked proteins such as CD59 (Fig. 14.13). The cardinal role of complement activation in PNH pathogenesis has been proven by therapeutic use of monoclonal antibodies that prevent the conversion of C5 to C5a. Such inhibitors not only reduce hemolysis and its attendant transfusion requirements, but also lower the risk of thrombosis by up to 90%. The drawbacks to C5 inhibitor therapy are high cost and an increased risk of serious or fatal meningococcal infection (as is true of individuals with inherited complement defects). Immunosuppressive drugs are sometimes beneficial for those with evidence of marrow aplasia. The only cure is hematopoietic stem cell transplantation, which is usually reserved for those with intercurrent aplastic anemia or myeloid malignancy.

Immunohemolytic Anemia

Immunohemolytic anemia is caused by antibodies that recognize red cells and cause their premature destruction. Although these disorders are commonly referred to as autoimmune hemolytic anemias, the designation immunohemolytic anemia is preferred because the immune reaction is initiated in some instances by an ingested drug. Immunohemolytic anemia can be classified based on the characteristics of the responsible antibody (Table 14.4).

The diagnosis of immunohemolytic anemia requires the detection of antibodies and/or complement on red cells from the patient. This is done using the *direct Coombs antiglobulin test*, in which the patient’s red cells are mixed with sera containing antibodies that are specific for human immunoglobulin (Ig) or complement. If either immunoglobulin or complement is present on the surface of the red cells, the antibodies cause agglutination, which is appreciated visually as clumping. In the *indirect Coombs antiglobulin test*, the patient’s serum is tested for its ability to agglutinate commercially available red cells bearing known antigens. This test is used to characterize the target antigen and temperature dependence of the responsible antibody. Quantitative immunologic tests to measure such antibodies directly are also available.

TABLE 14.4 Classification of Immunohemolytic Anemia

Warm Antibody Type (IgG Antibodies Active at 37°C)	
Primary (idiopathic)	
Secondary	Autoimmune disorders (particularly systemic lupus erythematosus)
	Drugs
	Lymphoid neoplasms
Cold Agglutinin Type (IgM Antibodies Active Below 37°C)	
Acute (mycoplasmal infection, infectious mononucleosis)	
Chronic	
	Idiopathic
	Lymphoid neoplasms
Cold Hemolysin Type (IgG Antibodies Active Below 37°C)	
Rare; occurs mainly in children following viral infections	

Warm Antibody Type. This form constitutes approximately 80% of cases of immunohemolytic anemia. It is caused by antibodies that bind stably to red cells at 37°C. About 50% of cases are idiopathic (primary); others are related to a predisposing condition (see Table 14.4) or exposure to a drug. Most causative antibodies are of the IgG class; less commonly, IgA antibodies are the culprits. The red cell hemolysis is mostly extravascular. IgG-coated red cells bind to Fc receptors on phagocytes, which remove red cell membrane during “partial” phagocytosis. As in hereditary spherocytosis, the loss of membrane converts the red cells to spherocytes, which are sequestered and destroyed in the spleen. Moderate splenomegaly due to hyperplasia of splenic phagocytes is usually seen.

As with other autoimmune disorders, the cause of primary immunohemolytic anemia is unknown. In cases that are idiopathic, the antibodies are directed against red cell surface proteins, often components of the Rh blood group complex. In drug-induced cases, two mechanisms have been described:

- **Antigenic drugs.** In this setting hemolysis usually follows large intravenous doses of the offending drug and occurs 1 to 2 weeks after therapy is initiated. These drugs, exemplified by penicillin and cephalosporins, bind to the red cell membrane and create a new antigenic determinant that is recognized by antibodies. The responsible antibodies sometimes fix complement and cause intravascular hemolysis, but more often they act as opsonins that promote extravascular hemolysis within phagocytes.
- **Tolerance-breaking drugs.** These drugs, of which the antihypertensive agent α -methyldopa is the prototype, break tolerance in some unknown manner, leading to the production of antibodies against red cell antigens, particularly the Rh blood group antigens. About 10% of patients taking α -methyldopa develop autoantibodies, as assessed by the direct Coombs test, and roughly 1% develop clinically significant hemolysis.

Treatment of warm antibody immunohemolytic anemia centers on the removal of initiating factors (i.e., drugs); when this is not feasible, immunosuppressive drugs and splenectomy are the mainstays.

Cold Agglutinin Type. This type of immunohemolytic anemia is caused by IgM antibodies that bind to red cells avidly at low temperatures (0°C to 4°C) but not at 37°C. It accounts for 15% to 20% of cases. Cold agglutinin antibodies sometimes appear transiently following certain infections, such as with *Mycoplasma pneumoniae*, Epstein-Barr virus, cytomegalovirus, influenza virus, and human immunodeficiency virus (HIV). In these settings, the disorder is self-limited and

the antibodies rarely induce clinically important hemolysis. Chronic cold agglutinin immunohemolytic anemia occurs in association with certain B-cell neoplasms or as an idiopathic condition.

Clinical symptoms result from binding of IgM to red cells in vascular beds where the temperature may fall below 30°C, such as in exposed fingers, toes, and ears. IgM binding agglutinates red cells and fixes complement rapidly. As the blood recirculates and warms, IgM is released, usually before complement-mediated hemolysis can occur; therefore, intravascular hemolysis is usually not seen. However, the transient interaction with IgM is sufficient to deposit sublytic quantities of C3b, an excellent opsonin, which leads to the removal of red cells by phagocytes in the spleen, liver, and bone marrow (extravascular hemolysis). The hemolysis is of variable severity. Vascular obstruction caused by agglutinated red cells may produce pallor, cyanosis, and Raynaud phenomenon (Chapter 11) in parts of the body that are exposed to cold temperatures. Chronic cold agglutinin immunohemolytic anemia caused by IgM antibodies may be difficult to treat. The best approach, when possible, is avoidance of cold temperatures.

Cold Hemolysin Type. Cold hemolysins are autoantibodies responsible for an unusual entity known as *paroxysmal cold hemoglobinuria*. This rare disorder may cause substantial, sometimes fatal, intravascular hemolysis and hemoglobinuria. The autoantibodies are IgGs that bind to the P blood group antigen on the red cell surface in cool, peripheral regions of the body. Complement-mediated lysis occurs when the cells recirculate to the body's warm core, where the complement cascade functions more efficiently. Most cases are seen in children following viral infections; in this setting the disorder is transient and most of those affected recover within 1 month.

Hemolytic Anemia Due to Red Cell Trauma

The most significant hemolysis caused by trauma to red cells is seen in individuals with cardiac valve prostheses and microangiopathic disorders. Artificial mechanical cardiac valves are more frequently implicated than are bioprosthetic porcine or bovine valves. The hemolysis stems from shear forces produced by turbulent blood flow and pressure gradients across damaged valves. *Microangiopathic hemolytic anemia* is most commonly seen with disseminated intravascular coagulation (DIC), but it also occurs in thrombotic thrombocytopenic purpura (TTP), hemolytic uremic syndrome (HUS), malignant hypertension, systemic lupus erythematosus, and disseminated cancer. The common pathogenic feature in these disorders is microvascular lesions that result in luminal narrowing, often due to the deposition of thrombi, producing shear stresses that mechanically injure passing red cells. Regardless of the cause, traumatic damage leads to intravascular hemolysis and the appearance of red cell fragments (*schistocytes*), “burr cells,” “helmet cells,” and “triangle cells” in blood smears (Fig. 14.14).

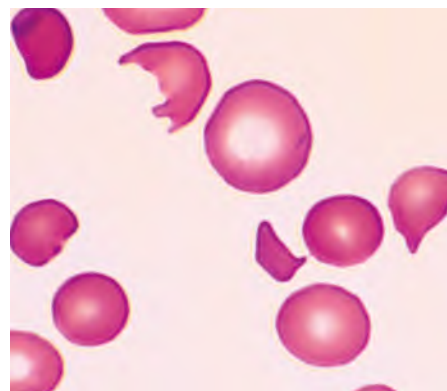


Fig. 14.14 Microangiopathic hemolytic anemia. A peripheral blood smear from a patient with hemolytic uremic syndrome shows several fragmented red cells. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Sickle Cell Anemia

- Autosomal recessive disorder resulting from a mutation in β -globin that causes deoxygenated hemoglobin to self-associate into long polymers that distort (sickle) the red cell
- Episodic lockage of vessels by sickle red cells causes pain crises and tissue infarction, particularly of the marrow and spleen
- Red cell membrane damage caused by repeated bouts of sickling results in a moderate to severe hemolytic anemia

Thalassemia

- Autosomal codominant disorder caused by mutations in α - or β -globin that reduce hemoglobin synthesis, resulting in a microcytic, hypochromic anemia
- In β -thalassemia, unpaired α -globin chains form aggregates that damage red cell precursors and further impair erythropoiesis
- Ineffective erythropoiesis increases iron absorption and can lead to systemic iron overload

Immunoheolytic Anemia

- Caused by antibodies against either normal red cell constituents or antigens modified by haptens (e.g., drugs)
- Antibody binding results in either red cell opsonization and extravascular hemolysis or (uncommonly) complement fixation and intravascular hemolysis

Anemias of Diminished Erythropoiesis

Although anemias caused by inadequate red cell production are heterogeneous, they can be classified into several major categories based on pathophysiology (see Table 14.1). The most common and important anemias associated with red cell underproduction are those caused by nutritional deficiencies, followed by those that arise secondary to chronic inflammation or renal failure. Also included are less common disorders that lead to generalized bone marrow failure, such as aplastic anemia, primary hematopoietic neoplasms (Chapter 13), and infiltrative disorders that replace the marrow (e.g., metastatic cancer and disseminated granulomatous disease). We first discuss the extrinsic causes of diminished erythropoiesis, which are more common and clinically important, and then nonneoplastic intrinsic causes.

Megaloblastic Anemia

The common theme among the various causes of megaloblastic anemia is an impairment of DNA synthesis that leads to ineffective hematopoiesis and distinctive morphologic changes, including

KEY CONCEPTS

Hereditary Spherocytosis

- Autosomal dominant disorder caused by mutations that affect the red cell membrane skeleton, leading to loss of membrane and eventual conversion of red cells to spherocytes, which are phagocytosed and removed in the spleen
- Manifested by anemia and splenomegaly

Glucose-6-Phosphate Dehydrogenase Deficiency

- X-linked disorder caused by mutations that destabilize G6PD, making red cells susceptible to oxidant damage

TABLE 14.5 Causes of Megaloblastic Anemia

Vitamin B₁₂ Deficiency
Decreased Intake
Inadequate diet, vegetarianism
Impaired Absorption
Intrinsic factor deficiency
Pernicious anemia
Gastrectomy
Malabsorption states
Diffuse intestinal disease (e.g., lymphoma, systemic sclerosis)
Ileal resection, ileitis
Competitive parasitic uptake
Fish tapeworm infestation
Bacterial overgrowth in blind loops and diverticula of bowel
Folic Acid Deficiency
Decreased Intake
Inadequate diet, alcoholism, infancy
Impaired Absorption
Malabsorption states
Intrinsic intestinal disease
Anticonvulsants, oral contraceptives
Increased Loss
Hemodialysis
Increased Requirement
Pregnancy, infancy, disseminated cancer, markedly increased hematopoiesis
Impaired Utilization
Folic acid antagonists
Unresponsive to Vitamin B₁₂ or Folic Acid Therapy
Metabolic inhibitors of DNA synthesis and/or folate metabolism (e.g., methotrexate)

Modified from Beck WS. Megaloblastic anemias. In: Wyngaarden JB, Smith LH, eds. *Cecil Textbook of Medicine*. 18th ed. WB Saunders; 1988:900.

abnormally large erythroid precursors and red cells. The causes of megaloblastic anemia are given in Table 14.5. The following discussion first describes the common features and then turns to the two principal subtypes: pernicious anemia (the major form of vitamin B₁₂ deficiency anemia) and folate deficiency anemia.

Some of the metabolic roles of vitamin B₁₂ and folate are considered later. For now, it suffices that vitamin B₁₂ and folic acid are cofactors required for the synthesis of thymidine, one of the four bases found in DNA. A deficiency of these vitamins or impairment of their metabolism results in defective nuclear maturation due to deranged or inadequate DNA synthesis, with an attendant delay or block in cell division.

MORPHOLOGY

Certain peripheral blood findings are common to all forms of megaloblastic anemia. The presence of red cells that are macrocytic and oval (**macro-ovalocytes**) is highly characteristic. Because they are larger than normal and contain ample hemoglobin, most macrocytes lack the central pallor of normal red cells and appear “hyperchromic,” but the mean cell hemoglobin concentration is not elevated. There is marked variation in red cell size (anisocytosis) and shape (poikilocytosis). The reticulocyte count is low. Nucleated red cell progenitors occasionally appear in the circulating blood when anemia is severe. Neutrophils are also larger than normal and show **nuclear hypersegmentation**, having five or more nuclear lobules instead of the normal three to four (Fig. 14.15).

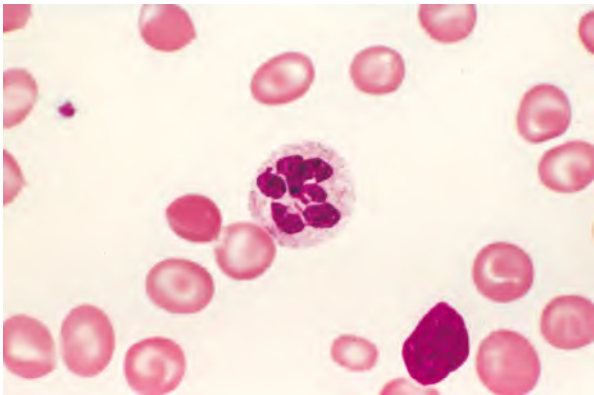


Fig. 14.15 Megaloblastic anemia. A peripheral blood smear shows a hypersegmented neutrophil with a six-lobed nucleus. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

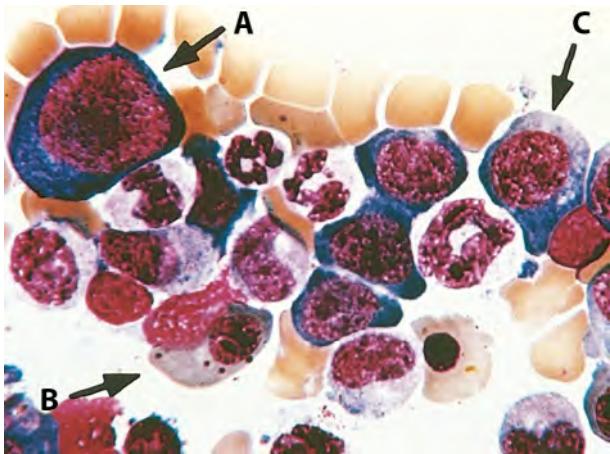


Fig. 14.16 Megaloblastic anemia (bone marrow aspirate). A to C, Megaloblasts in various stages of differentiation. Note that the orthochromatic megaloblast (B) is hemoglobinized (as revealed by cytoplasmic color), but in contrast to normal orthochromatic normoblasts, the nucleus is not pyknotic. The early erythroid precursors (A and C) and the granulocytic precursors also are enlarged and have abnormally immature chromatin. (Courtesy Dr. Jose Hernandez, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

The marrow is usually markedly hypercellular due to increased numbers of hematopoietic precursors. **Megaloblastic changes** are detected at all stages of erythroid development. The most primitive cells (promegaloblasts) are large and have deeply basophilic cytoplasm, prominent nucleoli, and a distinctive, fine nuclear chromatin pattern (Fig. 14.16). As these cells differentiate and begin to accumulate hemoglobin, the nucleus retains its finely distributed chromatin instead of developing the clumped pyknotic chromatin typical of normoblasts. Although nuclear maturation is delayed, cytoplasmic maturation and hemoglobin accumulation proceed at a normal pace, leading to nuclear-to-cytoplasmic asynchrony. Because DNA synthesis is impaired in all proliferating cells, granulocytic precursors also display dysmaturation in the form of **giant metamyelocytes** and **band forms**. Megakaryocytes may also be abnormally large and have bizarre, multilobate nuclei.

The marrow hyperplasia is a response to increased levels of growth factors such as erythropoietin. However, the derangement in DNA synthesis causes most precursors to undergo apoptosis in the marrow (ineffective hematopoiesis) and leads to pancytopenia. The anemia is further exacerbated by a mild degree of red cell hemolysis of uncertain etiology.

Anemia of Vitamin B₁₂ Deficiency: Pernicious Anemia

Pernicious anemia is a specific form of megaloblastic anemia caused by an autoimmune gastritis that impairs the production of intrinsic factor, which is required for vitamin B₁₂ uptake from the gut.

Normal Vitamin B₁₂ Metabolism. Vitamin B₁₂ is a complex organo-metallic compound also known as cobalamin that is present in animal products such as meat, fish, milk, and eggs. The daily requirement is 2 to 3 µg. A diet that includes animal products contains significantly more than the minimal daily requirement and normally results in the accumulation of intrahepatic stores of vitamin B₁₂ that are sufficient to last for several years. By contrast, plants and vegetables contain little cobalamin and strictly vegetarian or macrobiotic diets do not provide adequate amounts of this essential nutrient.

Absorption of vitamin B₁₂ requires intrinsic factor, which is secreted by the parietal cells of the fundic mucosa (Fig. 14.17). Vitamin B₁₂ is freed from binding proteins in food through the action of pepsin in the stomach and binds to a salivary protein called *haptocorrin*. In the duodenum, bound vitamin B₁₂ is released from haptocorrin by the action of pancreatic proteases, allowing it to associate with intrinsic factor. This complex is then transported to the ileum, where it binds a receptor for intrinsic factor called *cubilin* on ileal epithelial cells and is endocytosed. Within ileal cells, vitamin B₁₂ associates with its major carrier protein, transcobalamin II, and is secreted into the plasma. Transcobalamin II delivers vitamin B₁₂ to the liver and other cells of the body, including rapidly proliferating cells in the bone marrow and the gastrointestinal tract. In addition to this major pathway for uptake, B₁₂ can also be absorbed (albeit inefficiently) by passive diffusion, making it feasible to treat pernicious anemia with high doses of oral vitamin B₁₂.

Biochemical Functions of Vitamin B₁₂. Only two reactions in humans are known to require vitamin B₁₂. In one, methylcobalamin serves as a cofactor in the conversion of homocysteine to methionine by the enzyme methionine synthase (Fig. 14.18). In the process, methylcobalamin yields a methyl group that is recovered from N⁵-methyltetrahydrofolic acid (N⁵-methyl FH₄), the principal form of folic acid in plasma. In the same reaction, N⁵-methyl FH₄ is converted to tetrahydrofolic acid (FH₄). FH₄ is crucial because it is required (through its derivative N^{5,10}-methylene FH₄) for the conversion of deoxyuridine monophosphate (dUMP) to deoxythymidine monophosphate (dTMP), a building block for DNA. It is postulated that impaired DNA synthesis in vitamin B₁₂ deficiency stems from the reduced availability of FH₄, most of which is “trapped” as N⁵-methyl FH₄. The FH₄ deficit may be exacerbated by an “internal” deficiency of metabolically active polyglutamylated forms of FH₄. This stems from a requirement for vitamin B₁₂ in the synthesis of methionine (see Fig. 14.18), which contributes a carbon group that is needed in the metabolic reactions that create folate polyglutamates (Fig. 14.19). Whatever the mechanism, lack of folate is the proximate cause of anemia in vitamin B₁₂ deficiency, as the anemia improves following the administration of folic acid.

The neurologic complications associated with vitamin B₁₂ deficiency are more enigmatic because they are not improved (and may be worsened) by folate administration. The only other known reaction that depends on vitamin B₁₂ is the isomerization of methylmalonyl coenzyme A to succinyl coenzyme A by the enzyme methylmalonyl-coenzyme A mutase, which requires adenosylcobalamin. A deficiency of vitamin B₁₂ thus leads to increased plasma and urine levels of methylmalonic acid. Interruption of this reaction and the

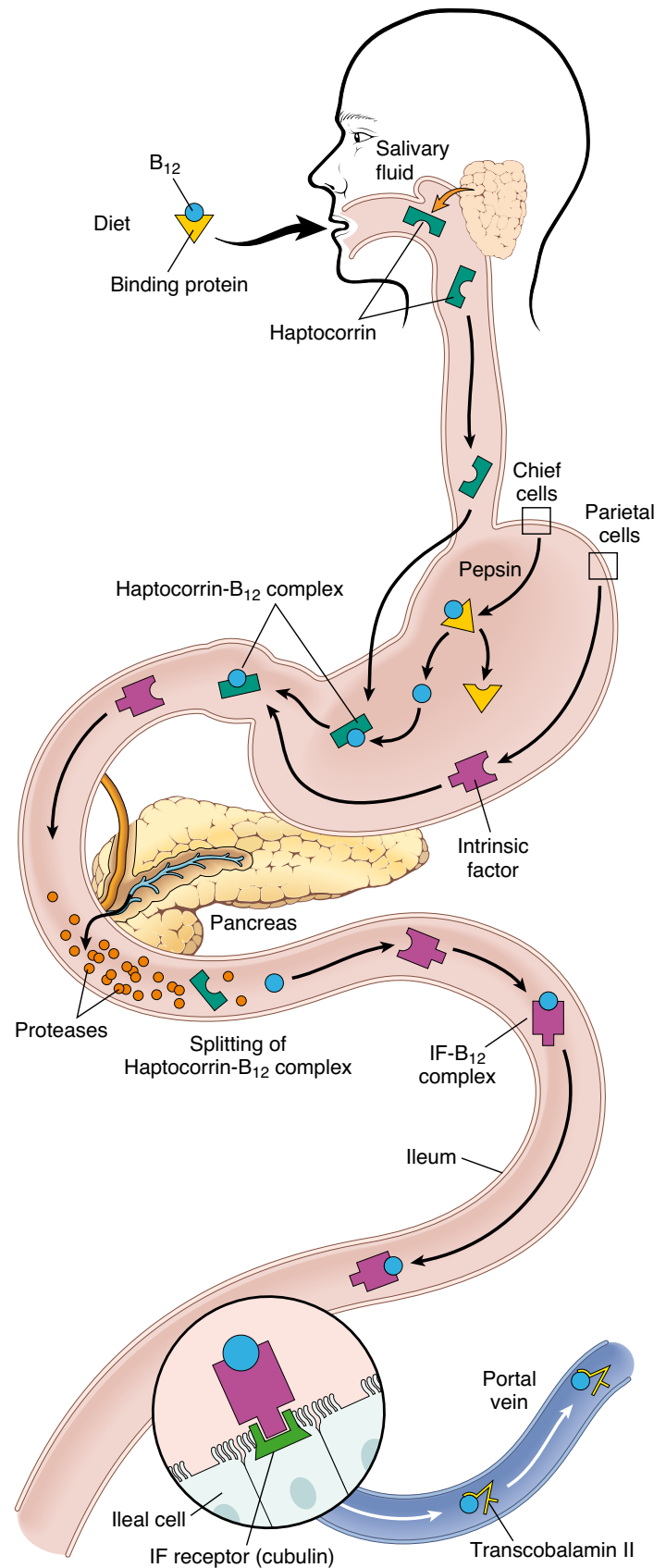


Fig. 14.17 Schematic illustration of vitamin B₁₂ absorption. IF, Intrinsic factor.

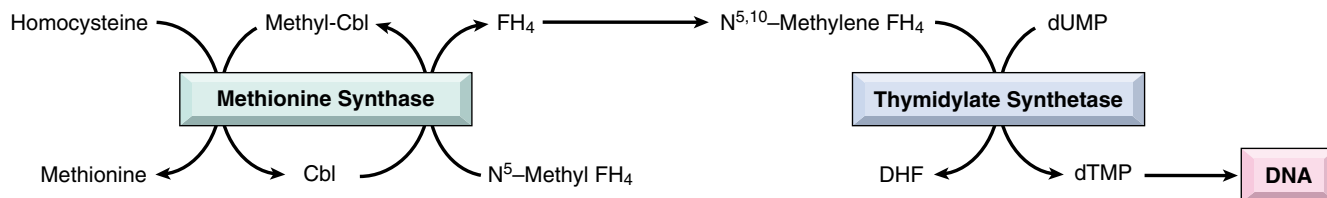


Fig. 14.18 Relationship of N^5 -methyl FH_4 , methionine synthase, and thymidylate synthetase. In cobalamin (Cbl) deficiency, folate is sequestered as N^5 -methyl FH_4 . This ultimately deprives thymidylate synthetase of its folate coenzyme ($N^{5,10}$ -methylene FH_4), thereby impairing DNA synthesis. *dTMP*, Deoxythymidine monophosphate; *dUMP*, deoxyuridine monophosphate; FH_4 , tetrahydrofolic acid.

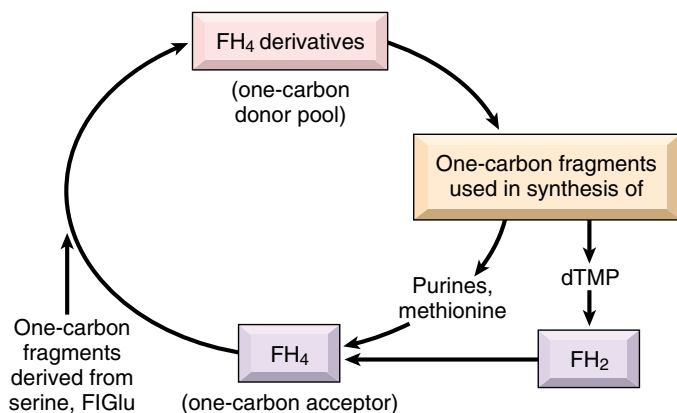


Fig. 14.19 Role of folate derivatives in the transfer of one-carbon fragments for synthesis of biologic macromolecules. *dTMP*, Deoxythymidine monophosphate; FH_4 , tetrahydrofolic acid; FH_2 , dihydrofolic acid; *FIGlu*, formiminoglutamate.

consequent buildup of methylmalonate and propionate (a precursor) could lead to the formation and incorporation of abnormal fatty acids into neuronal lipids. It has been suggested that this biochemical abnormality predisposes to myelin breakdown, producing *subacute combined degeneration of the spinal cord* (Chapter 28). However, rare individuals with hereditary deficiencies of methylmalonyl-coenzyme A mutase do not have these abnormalities, casting doubt on this explanation.

Although somewhat more prevalent in Scandinavian and other Northern European populations, pernicious anemia occurs in all groups, including people of African and Hispanic descent. It is a disease of older adults; the median age at diagnosis is 60 years and it is rare in people younger than 30 years. A genetic predisposition is strongly suspected, but no definable genetic pattern of transmission has been discerned. As described later, many affected individuals have a propensity to form autoantibodies.

Pathogenesis

Pernicious anemia is believed to result from an autoimmune attack on the gastric mucosa. Histologically, there is a chronic atrophic gastritis marked by a loss of parietal cells, a prominent infiltrate of lymphocytes and plasma cells, and megaloblastic changes in mucosal cells resembling those found in erythroid precursors. Three types of autoantibodies are present in many, but not all, patients. About 75% of patients have a type I antibody that blocks the binding of vitamin B_{12} to intrinsic factor. Type II antibodies block binding of intrinsic factor–vitamin B_{12} complexes to cubulin, preventing intestinal

absorption; these are also present in a large proportion of patients with pernicious anemia. Both type I and type II antibodies are found in plasma and gastric juice. Type III antibodies are present in 85% to 90% of patients and recognize the α and β subunits of the gastric proton pump, a component of the microvilli of gastric parietal cells; however, they are not specific, being found in as many as 50% of older adults with idiopathic chronic gastritis.

Autoantibodies are of diagnostic utility but are not thought to be the primary cause of the gastric pathology; rather, it seems that an autoreactive T-cell response initiates gastric mucosal injury and triggers the formation of autoantibodies. When the mass of intrinsic factor–secreting cells falls below a threshold (and reserves of stored vitamin B_{12} are depleted), anemia develops. Notably, pernicious anemia is associated with other autoimmune disorders, particularly autoimmune thyroiditis and adrenalitis, suggesting that it arises in individuals with a predisposition to develop autoimmunity.

Vitamin B_{12} deficiency may also arise from other causes, most of which impair absorption of the vitamin at one of the steps outlined earlier (see Table 14.5). With achlorhydria and loss of pepsin secretion (which occurs in some older adults), vitamin B_{12} is not readily released from proteins in food. With gastrectomy, intrinsic factor is lost. With insufficiency of the exocrine pancreas, vitamin B_{12} cannot be released from haptocorrin–vitamin B_{12} complexes. Ileal resection or diffuse ileal disease may prevent adequate absorption of intrinsic factor–vitamin B_{12} complex. Certain tapeworms (particularly those acquired by eating raw fish) compete with the host for B_{12} and can induce a deficiency state. In some settings, such as pregnancy, hyperthyroidism, disseminated cancer, and chronic infection, an increased demand for vitamin B_{12} may produce a relative deficiency, even with normal absorption. Recreational use of nitrous oxide may oxidize and inactivate vitamin B_{12} , leading to symptomatic deficiencies.

MORPHOLOGY

The findings in the bone marrow and blood in pernicious anemia resemble those described earlier for other forms of megaloblastic anemia. The stomach typically shows diffuse chronic gastritis (Chapter 17). The most characteristic alteration is **fundic gland atrophy**, affecting both chief cells and parietal cells, the latter being virtually absent. The glandular epithelium is replaced by mucus-secreting goblet cells like those lining the large intestine, a form of metaplasia referred to as **intestinalization**. Some of the affected cells and their nuclei may be twice normal in size, a “megaloblastic” change analogous to that seen in the marrow. With time, the tongue may take on a shiny, glazed, “beefy” appearance (**atrophic glossitis**). Because the gastric atrophy and metaplastic changes are due to autoimmunity, they persist following

parenteral administration of vitamin B₁₂, whereas the “megaloblastic” changes in the marrow and gut are readily reversible.

Central nervous system lesions are found in about three-quarters of cases of florid pernicious anemia but can also be seen in the absence of overt hematologic findings. The principal alterations involve the cord, where there is **demyelination of the dorsal and lateral spinal tracts**, sometimes followed by loss of axons. These changes may give rise to spastic paraparesis, sensory ataxia, and severe paresthesias in the lower limbs. Less frequently, degenerative changes occur in the ganglia of the posterior roots and in peripheral nerves (Chapter 28).

Clinical Features

Pernicious anemia is insidious, and the anemia is often quite severe by the time it comes to medical attention. The course is progressive unless halted by therapy.

The diagnosis is based on (1) a moderate to severe megaloblastic anemia, (2) leukopenia with hypersegmented granulocytes, (3) low serum vitamin B₁₂, and (4) elevated serum levels of homocysteine and methylmalonic acid. Serum antibodies to intrinsic factor are highly specific for pernicious anemia. The diagnosis is confirmed by an outpouring of reticulocytes and a rise in hematocrit levels beginning about 5 days after parenteral administration of vitamin B₁₂.

Persons with atrophy and metaplasia of the gastric mucosa due to pernicious anemia are at increased risk for gastric carcinoma (Chapter 17). With parenteral or high-dose oral vitamin B₁₂, the anemia is cured and the progression of the peripheral neurologic disease can be reversed or at least halted, but the changes in the gastric mucosa and the risk of carcinoma are unaffected.

Anemia of Folate Deficiency

A deficiency of folic acid (pteroylmonoglutamic acid) results in megaloblastic anemia having the same pathologic features as that caused by vitamin B₁₂ deficiency. FH₄ serves as an acceptor of one-carbon fragments from compounds such as serine and formiminoglutamic acid, and the FH₄ derivatives so generated in turn donate the acquired one-carbon fragments in reactions synthesizing various metabolites (see Fig. 14.19). FH₄, then, can be viewed as the biologic “middleman” in a series of swaps involving one-carbon moieties. The most important metabolic processes depending on such transfers are (1) purine synthesis; (2) the conversion of homocysteine to methionine, a reaction also requiring vitamin B₁₂; and (3) dTMP synthesis. In the first two reactions, FH₄ is regenerated from its one-carbon carrier derivatives and is available to accept another one-carbon moiety and reenter the donor pool. In the synthesis of dTMP, dihydrofolic acid (FH₂) is produced that must be reduced by dihydrofolate reductase for reentry into the FH₄ pool. The reductase step is significant because this enzyme is susceptible to inhibition by various drugs. Among the molecules whose synthesis is dependent on folates, dTMP is perhaps the most important biologically because it is required for DNA synthesis. It should be apparent from this discussion that **suppressed synthesis of DNA, the common denominator of folic acid and vitamin B₁₂ deficiency, is the immediate cause of megaloblastosis.**

Etiology and Pathogenesis

The three major causes of folic acid deficiency are (1) decreased intake, (2) increased requirements, and (3) impaired utilization (see Table 14.5). Humans depend on dietary sources for folic acid. Most normal diets contain ample amounts. The richest sources are green vegetables such as lettuce, spinach, asparagus, and broccoli. Certain fruits (e.g., lemons, bananas, melons) and animal sources (e.g., liver) contain lesser amounts. The folic acid in these foods is largely in the form of

polyglutamates. Although abundant in raw foods, polyglutamates are sensitive to heat; boiling, steaming, or frying food for 5 to 10 minutes destroys up to 95% of the folate content. Intestinal conjugases split the polyglutamates into monoglutamates that are absorbed in the proximal jejunum. During intestinal absorption they are modified to 5-methyltetrahydrofolate, the transport form of folate. The body's reserves of folate are relatively modest, and a deficiency can arise within weeks to months if intake is inadequate.

Decreased intake can result from either a nutritionally inadequate diet or impairment of intestinal absorption. A normal diet contains amounts of folate in excess of the minimal daily adult requirement. Inadequate dietary intake is almost invariably associated with grossly deficient diets, which are most frequently encountered in those with chronic alcohol use disorder, the indigent, and the very old. In those with chronic alcohol use disorder and cirrhosis, other mechanisms of folate deficiency such as trapping of folate within the liver, excessive urinary loss, and disordered folate metabolism have also been implicated. Under these circumstances, the megaloblastic anemia is often accompanied by general malnutrition and manifestations of other avitaminoses, including cheilosis, glossitis, and dermatitis. Malabsorption syndromes such as sprue may lead to inadequate folate absorption, as may infiltrative diseases of the small intestine (e.g., lymphoma). In addition, certain drugs, particularly the anticonvulsant phenytoin and oral contraceptives, interfere with absorption.

Despite normal intake of folic acid, a relative deficiency can be encountered when requirements are increased. Conditions in which this is seen include pregnancy, infancy, derangements associated with hyperactive hematopoiesis (e.g., chronic hemolytic anemia), and disseminated cancer. In all these circumstances, the demands of increased DNA synthesis render normal intake inadequate.

Folic acid antagonists, such as methotrexate, inhibit dihydrofolate reductase and lead to a deficiency of FH₄. Inhibition of folate metabolism affects all rapidly proliferating tissues, particularly the bone marrow and the gastrointestinal tract. Many chemotherapeutic drugs used in the treatment of cancer damage DNA or inhibit DNA synthesis through other mechanisms, and these also cause megaloblastic changes in rapidly dividing cells.

Clinical Features

Because the megaloblastic anemia that results from folic acid deficiency is identical to that encountered in vitamin B₁₂ deficiency, the diagnosis of folate deficiency must be made by demonstration of decreased serum or red cell folate levels. As in vitamin B₁₂ deficiency, serum homocysteine levels are increased, but methylmalonate concentrations are normal. Importantly, neurologic changes do not occur.

Although prompt hematologic response heralded by reticulocytosis follows the administration of folic acid, it should be remembered that the hematologic symptoms of vitamin B₁₂ deficiency anemia also respond to folate therapy. As mentioned earlier, folate does not prevent (and may even exacerbate) the neurologic deficits seen in vitamin B₁₂ deficiency states. It is thus essential to exclude vitamin B₁₂ deficiency as the cause of megaloblastic anemia before initiating therapy with folate.

Iron Deficiency Anemia

Deficiency of iron is the most common nutritional disorder in the world and results in clinical signs and symptoms that are mostly related to inadequate hemoglobin synthesis. Although the prevalence of iron deficiency anemia is higher in low-resource countries, this form of anemia is common in the United States, particularly in toddlers, adolescent girls, and females of childbearing age. The factors underlying iron deficiency differ somewhat in various population groups and can be best considered in the context of normal iron metabolism.

TABLE 14.6 Iron Distribution in Healthy Young Adults (mg)

Pool	Males	Females
Total	3450	2450
Functional		
Hemoglobin	2100	1750
Myoglobin	300	250
Enzymes	50	50
Storage		
Ferritin, hemosiderin	1000	400

Iron Metabolism. Iron balance is maintained by regulating the absorption of dietary iron in the proximal duodenum (described later). By contrast, there is no regulated pathway for iron excretion and daily losses are limited to the 1 to 2 mg found in shed mucosal and skin epithelial cells. The normal daily diet in the United States contains 10 to 20 mg of iron, mostly in the form of heme in animal products, with the remainder being inorganic iron in vegetables. About 20% of heme iron (in contrast with 1% to 2% of nonheme iron) is absorbable, so the average diet in the United States contains sufficient iron to balance fixed daily losses. By contrast, in low-resource parts of the world, iron intake is often inadequate to make up for normal daily losses. This is compounded in some lower-resource areas by the high burden of parasites, such as hookworm, that may be associated with significant blood loss.

In the United States, average total body iron content is as high as 6 g in healthy males but is only 2.5 g in healthy young females because of blood loss during menstruation and increased demands during pregnancy. Most of the total body iron exists in the functional compartment (Table 14.6), consisting of iron in hemoglobin, myoglobin, and iron-containing enzymes such as catalase and cytochromes. Most remaining iron is found in the storage compartment, in which it exists in two major protein-bound forms:

- **Ferritin** is a protein-iron complex found at the highest levels in hepatocytes and splenic and bone marrow macrophages. Iron bound to ferritin is tightly sequestered, which is important because free iron is highly toxic (Chapter 18). In individuals with normal iron stores, most storage iron is bound to ferritin. Cell turnover releases some ferritin into the plasma, and in general plasma ferritin levels correlate with body iron stores. In iron deficiency, serum ferritin is below 12 $\mu\text{g/L}$, whereas in iron overload values approaching 5000 $\mu\text{g/L}$ may be seen.
- **Hemosiderin granules** are created by the aggregation of ferritin following partial degradation within lysosomes. Iron in hemosiderin is chemically reactive and turns blue-black when exposed to potassium ferrocyanide, which is the basis for the *Prussian blue stain*. In iron-overloaded cells, most iron is stored as hemosiderin.

Of physiologic importance, storage iron can be readily mobilized if iron requirements increase, as may occur after loss of blood.

Transfer of iron between the functional and storage pools (Fig. 14.20) is mediated by a plasma iron-binding protein called *transferrin*, which is synthesized in the liver. In normal individuals, transferrin is about one-third saturated with iron, yielding serum iron levels that average 120 $\mu\text{g/dL}$ in males and 100 $\mu\text{g/dL}$ in females. The major function of transferrin is to deliver iron to cells, particularly erythroid precursors, which require iron to synthesize hemoglobin. Iron delivery is enabled by high-affinity receptors for transferrin on erythroid precursors that mediate iron import through receptor-mediated endocytosis.

The pathways responsible for regulation of iron absorption in the duodenum are now understood in reasonable detail (Fig. 14.21) and

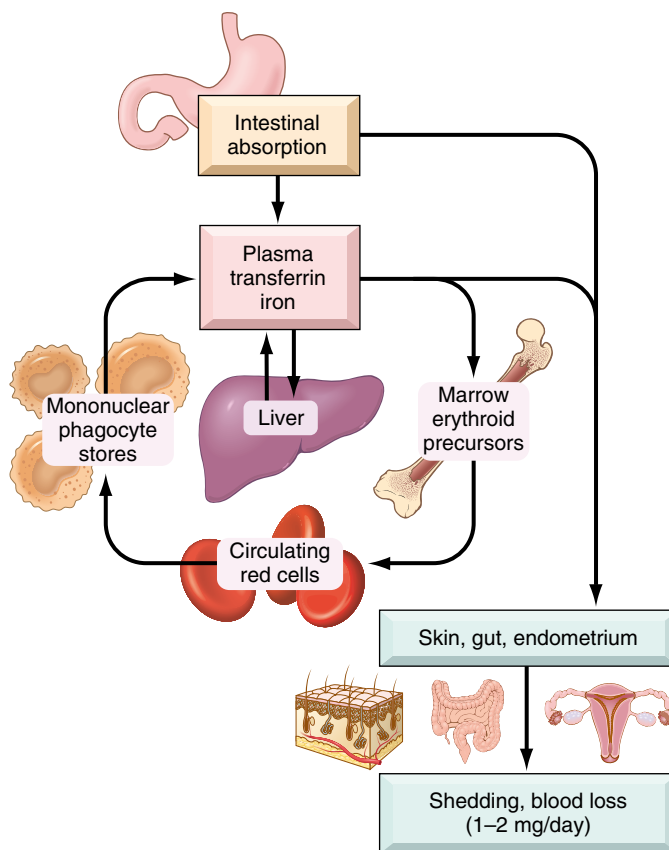


Fig. 14.20 Iron metabolism. Iron absorbed from the gut is bound to plasma transferrin and transported to the bone marrow, where it is delivered to developing red cells and incorporated into hemoglobin. Mature red cells are released into the circulation and, after 120 days, are ingested by macrophages, primarily in the spleen, liver, and bone marrow. Here iron is extracted from hemoglobin and recycled to plasma transferrin. At equilibrium, iron absorbed from the gut is balanced by losses in shed keratinocytes, enterocytes, and (in females) endometrium.

differ for nonheme and heme iron. Luminal nonheme iron is mostly in the Fe^{3+} (ferric) state and must first be reduced to Fe^{2+} (ferrous) iron by ferrireductases, such as b cytochromes. Fe^{2+} iron is then transported across the apical membrane by divalent metal transporter 1 (DMT1). Heme iron is moved across the apical membrane into the cytoplasm through transporters that are incompletely characterized. Here, it is metabolized to release Fe^{2+} iron, which is believed to enter a common pool with nonheme Fe^{2+} iron. As mentioned, heme iron is highly bioavailable, whereas the absorption of nonheme iron is variable and often inefficient, being inhibited by substances in the diet that bind and stabilize Fe^{3+} iron. Frequently, less than 5% of dietary nonheme iron is absorbed.

Once in duodenal cells, Fe^{2+} iron can follow one of two pathways: transport to the blood or storage as mucosal iron within ferritin. Fe^{2+} iron destined for the circulation is transported from the cytoplasm across the basolateral enterocyte membrane by ferroportin. This process is coupled to the oxidation of Fe^{2+} iron to Fe^{3+} iron, which is carried out by the iron oxidases hephaestin and ceruloplasmin. Newly absorbed Fe^{3+} iron binds rapidly to transferrin, which delivers iron to red cell progenitors in the marrow (see Fig. 14.20). Both DMT1 and ferroportin are widely distributed in the body and are involved in iron transport in other tissues as well. For example, DMT1 mediates the uptake of “functional” iron (derived from endocytosed transferrin)

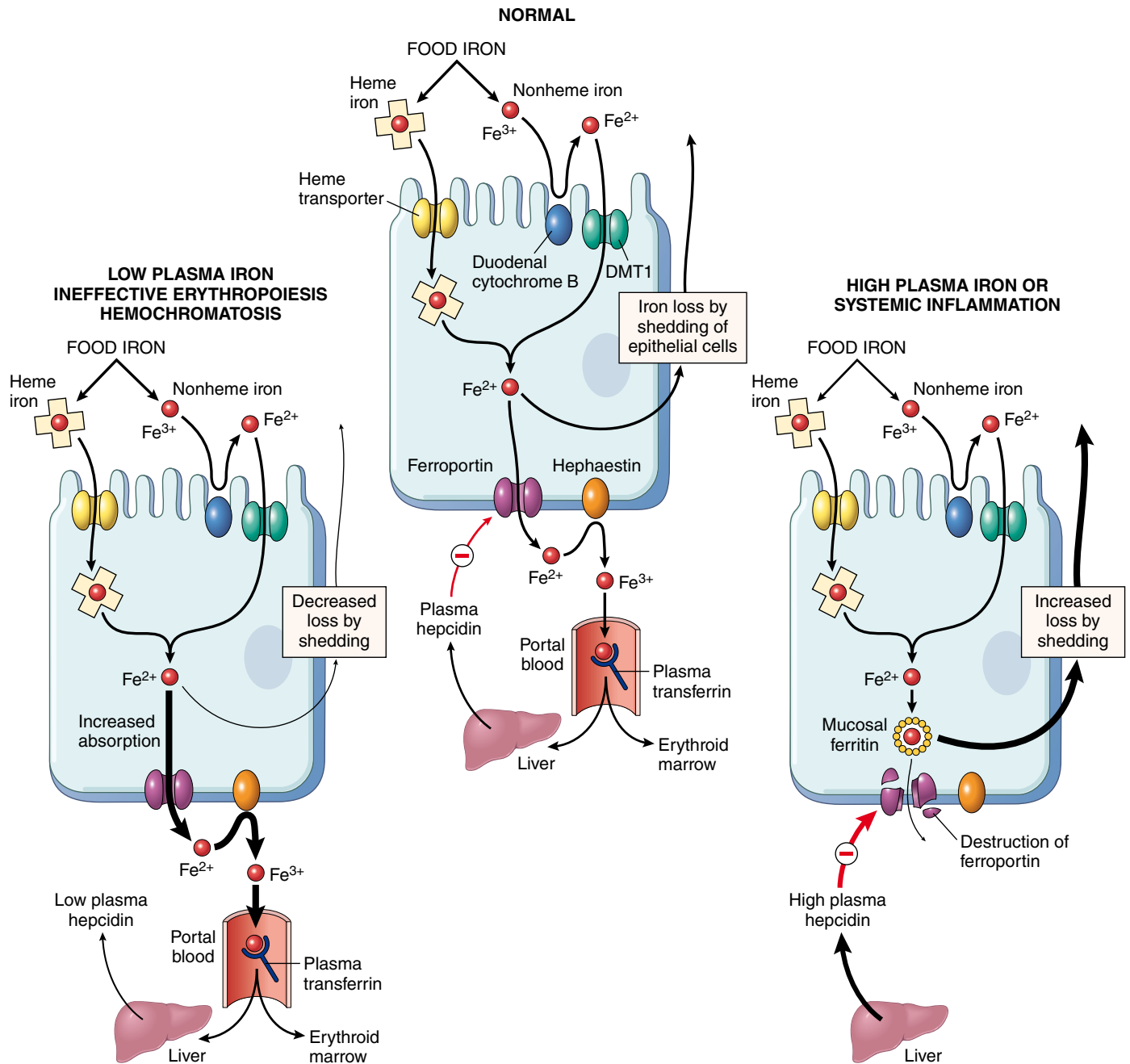


Fig. 14.21 Regulation of iron absorption. Duodenal epithelial cell uptake of heme and nonheme iron is depicted. When the storage sites of the body are replete with iron and erythropoietic activity is normal, plasma hepcidin balances iron uptake and loss to maintain iron hemostasis by downregulating ferroportin and limiting iron uptake (*middle*). Hepcidin rises in the setting of systemic inflammation or when iron levels are high, decreasing iron uptake by promoting the destruction of ferroportin (*right*), whereas it falls in the setting of low plasma iron or primary hemochromatosis, resulting in increased iron uptake (*left panel*). DMT1, Divalent metal transporter-1.

across lysosomal membranes into the cytosol of red cell precursors in the bone marrow, and ferroportin plays an important role in the release of storage iron from macrophages.

Iron absorption in the duodenum is regulated by hepcidin, a peptide that is synthesized and released from the liver in response to increases in intrahepatic iron levels. Hepcidin inhibits iron transfer from the enterocyte to plasma by binding to ferroportin, causing it to be endocytosed and degraded. As a result, as

hepcidin levels rise, iron becomes trapped within duodenal cells in the form of mucosal ferritin and is lost as these cells slough. Thus, when the body is replete with iron, high hepcidin levels inhibit its absorption into the blood. Conversely, with low body stores of iron, hepcidin levels fall, facilitating iron absorption. By inhibiting ferroportin, hepcidin not only reduces iron uptake from enterocytes but also suppresses iron release from macrophages, an important source of the iron that is used by erythroid precursors to make hemoglobin.

Alterations in hepcidin have a central role in diseases involving disturbances of iron metabolism. This is illustrated by the following examples.

- The *anemia of chronic inflammation* (described later) is caused in part by inflammatory mediators that increase hepatic hepcidin production, leading to sequestration of iron in macrophages.
- *Mutations that disable TMPRSS6*, a hepatic transmembrane serine protease that normally suppresses hepcidin production when iron stores are low, cause a rare form of microcytic anemia. Affected patients have high hepcidin levels, resulting in reduced iron absorption and failure to respond to iron therapy.
- Conversely, *hepcidin activity is inappropriately low in both primary and secondary systemic iron overload*, a condition known as *hemo-chromatosis*. As discussed in Chapter 18, the various inherited forms of primary hemochromatosis are caused by mutations in hepcidin, ferroportin, or the genes that regulate hepcidin expression.
- Secondary hemochromatosis can occur in diseases associated with ineffective erythropoiesis, such as β -thalassemia major (discussed earlier) and myelodysplastic neoplasms (Chapter 13), due to expansion of erythroid progenitors and release of increased amounts of erythroferrone, which inhibits hepatic hepcidin production.

Etiology and Pathogenesis

Iron deficiency can result from (1) dietary lack, (2) impaired absorption, (3) increased requirement, or (4) chronic blood loss. To maintain a normal iron balance, about 1 mg of iron must be absorbed from the diet every day. Because only 10% to 15% of ingested iron is absorbed, the daily iron requirement is 7 to 10 mg for adult males and 7 to 20 mg for adult females. Because the average daily dietary intake of iron in the United States is about 15 to 20 mg, most males ingest more than adequate iron, whereas many females consume marginal amounts. The bioavailability of dietary iron is as important as the overall content and is influenced by other dietary components. It is enhanced by ascorbic acid, citric acid, amino acids, and sugars in the diet, and it is inhibited by tannates (found in tea), carbonates, oxalates, and phosphates.

Dietary lack of iron is rare in high-resource countries, where on average about two-thirds of dietary iron is in the form of heme, mainly in meat. The situation is different in low-resource countries, where food is less abundant and most dietary iron is in the poorly absorbable inorganic form found in plants. However, dietary iron inadequacy occurs even in high-resource societies in the following groups:

- *Infants*, who are at high risk due to the very small amounts of iron in milk. Human breast milk provides only about 0.3 mg/L of iron. Cow's milk contains about twice as much iron, but its bioavailability is poor.
- *The impoverished*, who can have suboptimal diets for socioeconomic reasons at any age.
- *Older adults*, who often have restricted diets with little meat because of limited income or poor dentition.
- *Vegans* of any age, due to the lack of heme iron in their diets.

Impaired absorption is found in sprue, other causes of fat malabsorption (steatorrhea), and chronic diarrhea. Gastrectomy diminishes iron absorption by decreasing the acidity of the proximal duodenum (acidity enhances uptake) and by increasing the speed with which gut contents pass through the duodenum. Specific items in the diet, as is evident from the preceding discussion, can also affect absorption.

Increased requirement is an important cause of iron deficiency in growing infants, children, and adolescents, as well as premenopausal females, particularly during pregnancy. Females in low-resource countries having multiple, closely spaced pregnancies are at exceptionally high risk.

Chronic blood loss is the most common cause of iron deficiency in developed societies. External hemorrhage or bleeding into the gastrointestinal, urinary, or genital tracts depletes iron reserves. Iron deficiency in adult males and postmenopausal females in high-income countries must be attributed to gastrointestinal blood loss until proven otherwise. To prematurely ascribe iron deficiency in such individuals to any other cause is to run the risk of missing a gastrointestinal cancer or other bleeding lesion. An alert clinician investigating unexplained iron deficiency anemia occasionally discovers an occult bleeding source such as cancer and thereby saves a life.

Whatever its basis, iron deficiency leads to inadequate hemoglobin production and hypochromic microcytic anemia. At the outset of chronic blood loss or other states of negative iron balance, the iron storage pool may be adequate to maintain normal hemoglobin and hematocrit levels as well as normal serum iron and transferrin saturation. Progressive depletion of these reserves first lowers serum iron and transferrin saturation levels without producing anemia. In this early stage, there is increased erythroid activity in the bone marrow. Anemia appears only when iron stores are completely depleted and is accompanied by low levels of serum iron and ferritin and low transferrin saturation.

MORPHOLOGY

The bone marrow reveals a mild to moderate increase in erythroid progenitors. A diagnostically significant finding is the **absence of stainable iron in macrophages**, which is best assessed by performing Prussian blue stains on smears of aspirated marrow. In peripheral blood smears, the red cells are small (**microcytic**) and pale (**hypochromic**). Normal red cells with sufficient hemoglobin have a zone of central pallor measuring about one-third of the cell diameter. In established iron deficiency, the zone of pallor is enlarged; hemoglobin may be seen only in a narrow peripheral rim (Fig. 14.22). Poikilocytosis in the form of small, elongated red cells ("**pencil**" cells) is also characteristically seen.

Clinical Features

The clinical manifestations of the anemia are nonspecific and were detailed earlier. The dominating signs and symptoms frequently relate to the underlying cause, for example, gastrointestinal or gynecologic disease, malnutrition, pregnancy, or malabsorption. In severe long-standing iron

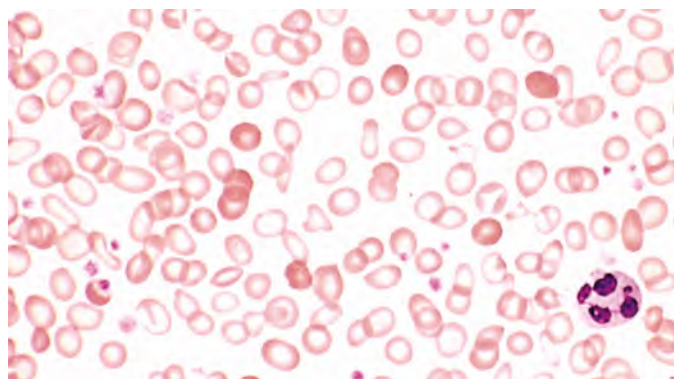


Fig. 14.22 Iron deficiency anemia (peripheral blood smear). Note the hypochromic microcytic red cells containing a narrow rim of peripheral hemoglobin. Scattered fully hemoglobinized cells, present due to recent blood transfusion, stand in contrast. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

deficiency, depletion of iron-containing enzymes in cells throughout the body also causes other changes, including koilonychia, alopecia, atrophic changes in the tongue and gastric mucosa, and intestinal malabsorption. Depletion of iron from the central nervous system may lead to the appearance of pica, in which affected individuals have a craving for non-foodstuffs such as clay, or food ingredients such as flour or ice (*pagophagia*). Pica is also seen in association with developmental disorders such as autism (in the absence of iron deficiency). Esophageal webs may appear together with microcytic hypochromic anemia and atrophic glossitis to complete the triad of findings in the rare *Plummer-Vinson syndrome* (Chapter 17).

The diagnosis of iron deficiency anemia ultimately rests on laboratory studies. Both the hemoglobin and hematocrit are depressed, usually to a moderate degree, in association with hypochromia, microcytosis, and modest poikilocytosis. The serum iron and ferritin are low, and the total plasma iron-binding capacity (reflecting elevated transferrin levels) is high. Low serum iron with increased iron-binding capacity results in a reduction of transferrin saturation to below 15%. Reduced iron stores inhibit hepcidin synthesis, and its serum levels fall. In uncomplicated cases, oral iron supplementation produces an increase in reticulocytes in about 5 to 7 days that is followed by a steady increase in blood counts and the normalization of red cell indices.

Anemia of Chronic Inflammation

This common form of anemia stems from a reduction in the proliferation of erythroid progenitors and impaired iron utilization because of inflammatory mediators. The chronic illnesses associated with this form of anemia can be grouped into three categories:

- *Chronic microbial infections*, such as osteomyelitis, bacterial endocarditis, and lung abscess
- *Chronic immune disorders*, such as rheumatoid arthritis and inflammatory bowel disease
- *Neoplasms*, such as carcinomas of the lung and breast, and Hodgkin lymphoma

Anemia of chronic inflammation is associated with low serum iron, reduced total iron-binding capacity, elevated plasma ferritin, and increased storage iron in tissue macrophages. Several effects of inflammation contribute to the observed abnormalities. Most notably, certain inflammatory mediators, particularly interleukin-6 (IL-6), stimulate an increase in the hepatic production of hepcidin. As discussed earlier, hepcidin inhibits ferroportin function in macrophages and reduces the transfer of iron from the storage pool to developing erythroid precursors in the bone marrow. As a result, erythroid precursors are starved for iron in the midst of plenty. In addition, these progenitors do not proliferate adequately because erythropoietin levels are inappropriately low for the degree of anemia. The precise mechanism underlying the reduction in erythropoietin is uncertain, but direct suppression of renal erythropoietin production by inflammatory cytokines is suspected.

Why does the body sequester iron in the setting of inflammation? The best guess is that it enhances the body's ability to fend off certain infections, particularly those caused by bacteria (e.g., *H. influenzae*) that require iron for pathogenicity. In this regard it is interesting to consider that hepcidin is structurally related to defensins, a family of peptides that have intrinsic antibacterial activity. This connection highlights the poorly understood but intriguing relationship between inflammation, innate immunity, and iron metabolism.

The anemia is usually mild and the dominant symptoms are those of the underlying disease. The red cells may be normocytic and normochromic, or slightly hypochromic and microcytic, as in anemia of iron deficiency. The presence of increased storage iron in marrow

macrophages, a high serum ferritin level, and a reduced total iron-binding capacity readily rule out iron deficiency as the cause of anemia. Only successful treatment of the underlying condition reliably corrects the anemia, but some patients, particularly those with cancer, benefit from administration of erythropoietin.

Aplastic Anemia

Aplastic anemia refers to a syndrome of chronic primary hematopoietic failure and attendant pancytopenia (anemia, neutropenia, and thrombocytopenia). In the majority of patients, autoimmune mechanisms are suspected, but inherited or acquired abnormalities of hematopoietic stem cells also contribute in a subset of patients.

Etiology and Pathogenesis

The most common associations with aplastic anemia are listed in Table 14.7. Most cases of “known” etiology follow exposure to chemicals and drugs. Some of the associated agents (including many cancer chemotherapy drugs and the organic solvent benzene) cause marrow suppression that is dose related and reversible. In other instances, aplastic anemia arises in an unpredictable, idiosyncratic fashion following exposure to drugs that normally cause little or no marrow suppression. The implicated drugs in these idiosyncratic reactions include chloramphenicol and gold salts.

Persistent marrow aplasia may also appear after a variety of viral infections, most commonly viral hepatitis, which is associated with approximately 5% of cases. Why aplastic anemia develops in certain individuals following infection is not understood.

Whole-body irradiation can destroy hematopoietic stem cells in a dose-dependent fashion. Persons who receive therapeutic irradiation or are exposed to radiation in nuclear accidents (e.g., Chernobyl) are at risk for marrow aplasia.

TABLE 14.7 Major Causes of Aplastic Anemia

Acquired
Idiopathic
Acquired stem cell defects
Immune mediated
Chemical Agents
Dose related
Alkylating agents
Antimetabolites
Benzene
Chloramphenicol
Inorganic arsenicals
Idiosyncratic
Chloramphenicol
Phenylbutazone
Organic arsenicals
Methylphenylethylhydantoin
Carbamazepine
Penicillamine
Gold salts
Physical Agents
Whole-body irradiation
Viral infections
Hepatitis (unknown virus)
Cytomegalovirus infections
Epstein-Barr virus infections
Herpes zoster (varicella zoster)
Inherited
Fanconi anemia
Telomerase defects

Specific abnormalities underlying some cases of aplastic anemia are:

- *Fanconi anemia* is a rare autosomal recessive disorder caused by defects in a multiprotein complex that is required for DNA repair (Chapter 7). Marrow hypofunction becomes evident early in life and is often accompanied by multiple congenital anomalies, such as kidney and spleen hypoplasia and bone anomalies, most commonly involving the thumbs or radii.
- Inherited defects in *telomerase* are found in 5% to 10% of adult-onset aplastic anemia. Telomerase is required for cellular immortality and limitless replication (Chapters 1 and 7). It might be anticipated, therefore, that deficits in telomerase activity could result in premature hematopoietic stem cell exhaustion and marrow aplasia. Even more common than telomerase mutations are abnormally short telomeres, which are found in the marrow cells of as many as one-half of those affected with aplastic anemia. It is unknown whether this shortening is due to other unappreciated telomerase defects or is a consequence of excessive stem cell replication.
- In most instances, however, no initiating factor can be identified; about 65% of cases fall into this idiopathic category.

Two major etiologies have been evoked to explain the pathogenesis of aplastic anemia, an extrinsic, immune-mediated suppression of marrow progenitors, and an intrinsic abnormality of stem cells (Fig. 14.23). Experimental studies have focused on a model in which activated T cells suppress hematopoietic stem cells. Stem cells may first be antigenically altered by exposure to drugs, infectious agents, or other unidentified environmental insults. This provokes a cellular immune response, during which activated Th1 cells produce cytokines such as interferon- γ

(IFN- γ) that suppress and kill hematopoietic progenitors. This scenario is supported by several observations.

- Analysis of the few remaining marrow stem cells from aplastic anemia marrows has revealed that genes involved in apoptosis and death pathways are up-regulated; of note, the same genes are up-regulated in normal stem cells exposed to interferon- γ .
- Even more compelling (and clinically relevant) evidence comes from experience with immunosuppressive therapy. Antithymocyte globulin and other immunosuppressive drugs such as cyclosporine improve marrow function in 60% to 70% of patients. It is proposed that these therapies work by suppressing or killing autoreactive T-cell clones. The antigens recognized by the autoreactive T cells are not well defined. In some instances, GPI-linked proteins may be the targets, possibly explaining the previously noted association of aplastic anemia and PNH.

Alternatively, the notion that aplastic anemia results from a fundamental stem cell abnormality is supported by: (1) the presence of karyotypic aberrations and acquired mutations involving cancer genes in many cases; (2) the occasional transformation of aplasia into a myeloid neoplasm, typically myelodysplastic neoplasm or acute myeloid leukemia; and the association with abnormally short telomeres. Some marrow insult (or a predisposition to DNA damage) presumably results in sufficient injury to limit the proliferative and differentiation capacity of stem cells. If the damage is extensive enough, aplastic anemia results. These two mechanisms are not mutually exclusive, because genetically altered stem cells might also express “neoantigens” that could serve as targets for a T-cell attack.

MORPHOLOGY

The markedly hypocellular bone marrow is largely devoid of hematopoietic cells; often only fat cells, fibrous stroma, and scattered lymphocytes and plasma cells remain. Marrow aspirates often yield little material (a “dry tap”); hence, aplasia is best appreciated in marrow biopsies (Fig. 14.24). Other nonspecific pathologic changes are related to granulocytopenia and thrombocytopenia, such as mucocutaneous bacterial infections and abnormal bleeding, respectively. If the anemia necessitates multiple transfusions, systemic hemosiderosis can appear.

Clinical Features

Aplastic anemia can occur at any age and in either sex. The onset is usually insidious. Initial manifestations vary depending on which cell line is predominantly affected, but pancytopenia inevitably appears, with the expected consequences. Anemia leads to progressive weakness, pallor, and dyspnea; thrombocytopenia is heralded by petechiae and ecchymoses; and neutropenia manifests as frequent and persistent infections or the sudden onset of chills, fever, and prostration. Splenomegaly is characteristically absent; if present, the diagnosis of aplastic anemia should be questioned. The red cells are usually slightly macrocytic and normochromic. Reticulocytopenia is the rule.

The diagnosis rests on examination of a bone marrow biopsy. It is important to distinguish aplastic anemia from other causes of pancytopenia, such as “aleukemic” leukemia and myelodysplastic neoplasm (Chapter 13), which may have identical clinical manifestations. In aplastic anemia, the marrow is hypocellular (usually markedly so), whereas myeloid neoplasms are usually associated with hypercellular marrows filled with neoplastic progenitors.

The prognosis is variable. Stem cell transplantation is the treatment of choice in those with a suitable donor and provides a 5-year survival of more than 75%. Older patients or those without suitable donors often respond well to immunosuppressive therapy.

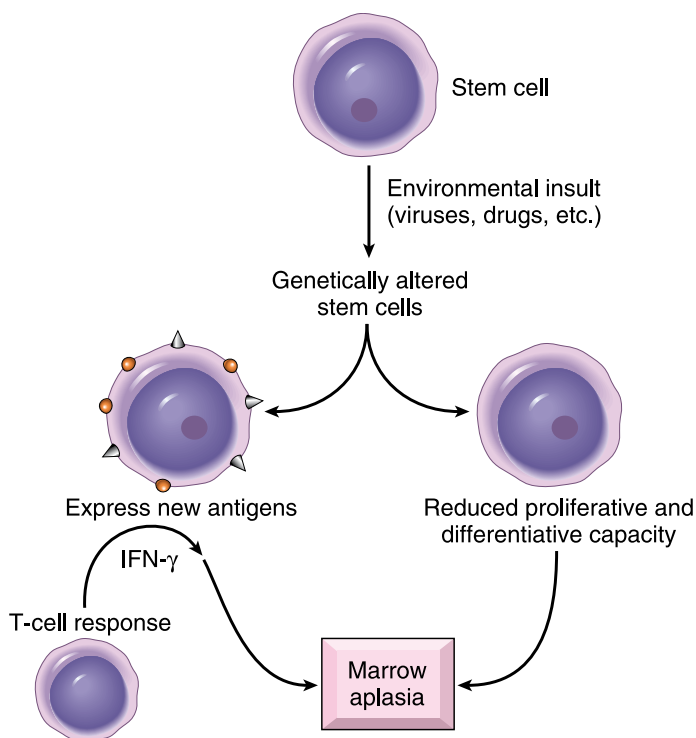


Fig. 14.23 Pathophysiology of aplastic anemia. Damaged stem cells can produce progeny expressing neoantigens that evoke an autoimmune reaction or give rise to a clonal population with reduced proliferative capacity. Either pathway could lead to marrow aplasia. IFN- γ , Interferon- γ .

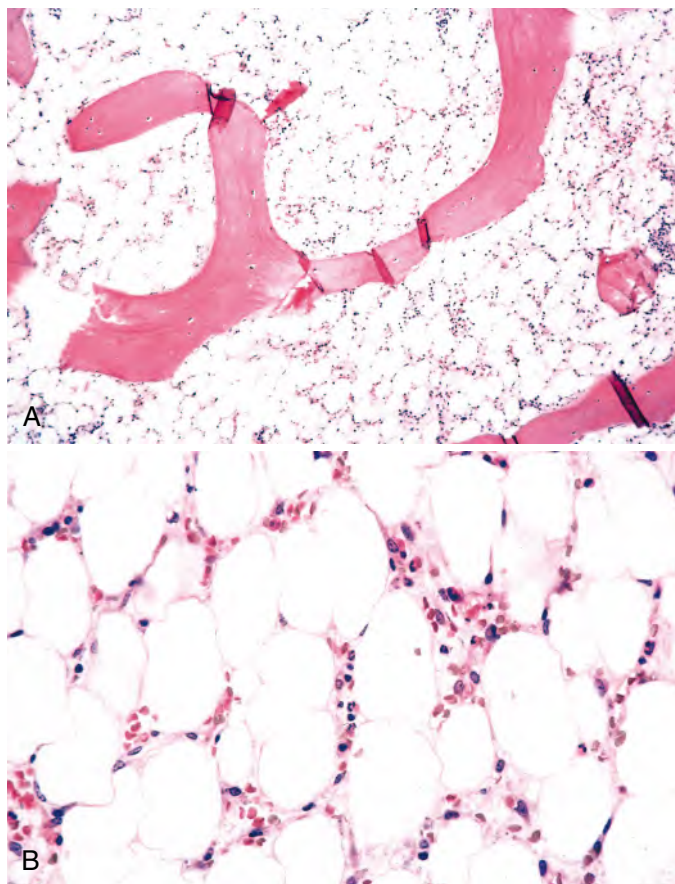


Fig. 14.24 Aplastic anemia (bone marrow biopsy). Markedly hypocellular marrow contains mainly fat cells. (A) Low power. (B) High power. (Courtesy Dr. Steven Kroft, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Pure Red Cell Aplasia

Pure red cell aplasia is a primary marrow disorder in which only erythroid progenitors are suppressed. In severe cases, red cell progenitors are completely absent from the marrow. It may occur in association with neoplasms, particularly thymoma and large granular lymphocytic leukemia (Chapter 13), drug exposures, autoimmune disorders, and parvovirus infection (see later). Except for those associated with parvovirus infection, most cases likely have an autoimmune basis. When a thymoma is present, resection leads to hematologic improvement in about one-half of the patients. In patients without thymoma, immunosuppressive therapy is often beneficial. Plasmapheresis may be helpful in unusual patients with neutralizing antibodies to erythropoietin, which may appear *de novo* or following the administration of recombinant erythropoietin.

A special form of red cell aplasia occurs in individuals infected with parvovirus B19, which preferentially infects and destroys red cell progenitors. Normal individuals clear parvovirus infections within 1 to 2 weeks; as a result, the aplasia is transient and clinically unimportant. However, as mentioned earlier, in persons with moderate to severe hemolytic anemias, even a brief cessation of erythropoiesis results in rapid worsening of the anemia, producing an aplastic crisis. In those who are severely immunosuppressed (e.g., persons with advanced HIV infection), an ineffective immune response sometimes permits the infection to persist, leading to chronic red cell aplasia and moderate to severe anemia.

Other Forms of Marrow Failure

Myelophthitic anemia describes a form of marrow failure in which space-occupying lesions replace normal marrow elements. The most common cause is metastatic cancer, most often carcinoma arising in the breast, lung, and prostate. However, any infiltrative process (e.g., granulomatous disease) involving the marrow can produce identical findings. Myelophthitic anemia is also a feature of the spent phase of myeloproliferative neoplasms (Chapter 13). All the responsible diseases cause marrow distortion and fibrosis, which displace normal marrow elements and disturb mechanisms that regulate the egress of red cells and granulocytes from the marrow. The latter effect causes the premature release of nucleated erythroid precursors and immature granulocytic forms into the circulation (*leukoerythroblastosis*) and the appearance of *teardrop-shaped red cells*, which are believed to be deformed during their tortuous escape from the fibrotic marrow.

Chronic renal failure, whatever its cause, is almost invariably associated with an anemia that tends to be roughly proportional to the severity of the uremia. The basis of anemia in renal failure is multifactorial, but the dominant cause is the diminished synthesis of erythropoietin by the damaged kidneys, leading to inadequate red cell production. Uremia also reduces red cell lifespan and impairs platelet function (both through uncertain mechanisms), and these effects may also contribute to anemia through extravascular hemolysis, abnormal bleeding, and eventually iron deficiency. Administration of recombinant erythropoietin and iron replacement therapy significantly improve the anemia of renal failure.

Hepatocellular disease, whether toxic, infectious, or cirrhotic, is associated with anemia attributed to decreased marrow function. Folate and iron deficiencies caused by poor nutrition and excessive bleeding often exacerbate anemia in this setting. Erythroid progenitors are preferentially affected; depression of the white cell and platelet counts also occurs, but less frequently. The anemia is often slightly macrocytic due to lipid abnormalities associated with liver failure, which cause red cell membranes to acquire phospholipid and cholesterol as they circulate in the peripheral blood, thereby increasing cell size.

Endocrine disorders, particularly hypothyroidism, may also be associated with a mild normochromic, normocytic anemia.

KEY CONCEPTS

Megaloblastic Anemia

- Caused by deficiencies of folate or vitamin B₁₂ that lead to inadequate synthesis of thymidine and defective DNA replication
- Results in enlarged abnormal hematopoietic precursors (megaloblasts), ineffective hematopoiesis, macrocytic anemia, and (in most cases) pancytopenia
- B₁₂ deficiency also associated with neurologic damage, particularly in the posterior and lateral tracts of the spinal cord

Iron Deficiency Anemia

- Caused by chronic bleeding or inadequate iron intake; results in insufficient hemoglobin synthesis and hypochromic, microcytic red cells

Anemia of Chronic Inflammation

- Caused by inflammatory cytokines, which increase hepcidin levels, thereby reducing iron absorption and sequestering iron in macrophages; also suppress erythropoietin production

Aplastic Anemia

- Caused by bone marrow failure (hypocellularity) due to diverse causes, including exposures to toxins and radiation, idiosyncratic reactions to drugs and viruses, and inherited defects in telomerase and DNA repair

Pure Red Cell Aplasia

- Parvovirus B19 infection: causes transient severe red cell aplasia in individuals with hemolytic anemia and chronic red cell aplasia in individuals who cannot clear the infection
- Also associated with thymoma, large granular lymphocytic leukemia, presence of neutralizing antibodies against erythropoietin, and other autoimmune phenomena

Other Causes of Underproduction Anemias

- Marrow replacement (tumors, granulomatous disease; so-called myelophthisic anemia), renal failure, endocrine disorders, liver failure

POLYCYTHEMIA

Polycythemia denotes an abnormally high number of circulating red cells, usually with a corresponding increase in the hemoglobin level; it may be relative (when there is hemoconcentration due to decreased plasma volume) or absolute (when there is an increase in the total red cell mass). Relative polycythemia results from dehydration, such as occurs with deprivation of water, prolonged vomiting or diarrhea, or excessive use of diuretics. Absolute polycythemia is *primary* when it results from an intrinsic abnormality of hematopoietic precursors and *secondary* when it stems from the response of red cell progenitors to elevated levels of erythropoietin. A pathophysiologic classification of polycythemia divided along these lines is given in Table 14.8.

The most common cause of primary polycythemia is *polycythemia vera*, a myeloproliferative neoplasm associated with mutations that lead to erythropoietin-independent growth of red cell progenitors (Chapter 13). Much less commonly, primary polycythemia results from familial erythropoietin receptor mutations that induce erythropoietin-independent receptor activation. One such individual won Olympic gold medals in cross-country skiing, having benefited from this natural form of blood doping! Secondary polycythemia stems from compensatory or pathologic increases in erythropoietin secretion. Causes of the latter include erythropoietin-secreting tumors and rare (but illustrative) inherited defects in various components of the renal oxygen-sensing pathway. These defects stabilize hypoxia-inducible factor (HIF), a transcription factor that stimulates the transcription of the erythropoietin gene.

TABLE 14.8 Pathophysiologic Classification of Polycythemia

Relative
Reduced plasma volume (hemoconcentration)
Absolute
Primary (Low Erythropoietin)
Polycythemia vera
Inherited erythropoietin receptor mutations (rare)
Secondary (High Erythropoietin)
Compensatory
Lung disease
High-altitude living
Cyanotic heart disease
Paraneoplastic
Erythropoietin-secreting tumors (e.g., renal cell carcinoma, hepatocellular carcinoma, cerebellar hemangioblastoma)
Hemoglobin mutants with high oxygen affinity
Inherited defects that stabilize HIF-1α
Chuvash polycythemia (homozygous <i>VHL</i> mutations)
Prolyl hydroxylase mutations

HIF-1α, Hypoxia-inducible factor 1α.

BLEEDING DISORDERS: HEMORRHAGIC DIATHESSES

Excessive bleeding can result from (1) increased fragility of vessels, (2) platelet deficiency or dysfunction, and (3) derangement of coagulation, alone or in combination. Before discussing specific bleeding disorders, it is helpful to review the common laboratory tests used in the evaluation of abnormal bleeding. The normal hemostatic response involves the blood vessel wall, the platelets, and the clotting cascade (Chapter 4). The following tests are used to evaluate different aspects of hemostasis:

- *Prothrombin time (PT)*. This test assesses the extrinsic and common coagulation pathways. The clotting of plasma after addition of an exogenous source of tissue thromboplastin (e.g., brain extract) and Ca²⁺ ions is measured in seconds. A prolonged PT can result from deficiency or dysfunction of factor V, factor VII, factor X, prothrombin, or fibrinogen.
- *Partial thromboplastin time (PTT)*. This test assesses the intrinsic and common clotting pathways. The clotting of plasma after addition of kaolin, cephalin, and Ca²⁺ ions is measured in seconds. Kaolin activates the contact-dependent factor XII and cephalin substitutes for platelet phospholipids. Prolongation of the PTT can be due to deficiency or dysfunction of factors V, VIII, IX, X, XI, or XII, prothrombin, or fibrinogen, or to interfering antiphospholipid antibodies (Chapter 4).
- *Platelet counts*. These are obtained on anticoagulated blood using an electronic particle counter. The reference range is 150 × 10³ to 350 × 10³ platelets per μL. Abnormal platelet counts should be confirmed by inspection of a peripheral blood smear, as clumping of platelets during automated counting can cause spurious “thrombocytopenia.” High counts may be indicative of a myeloproliferative neoplasm, such as essential thrombocythemia (Chapter 13), but are more likely to reflect reactive processes that increase platelet production (e.g., systemic inflammation).
- *Tests of platelet function*. At present, no single test provides an adequate assessment of the complex functions of platelets. Specialized tests that are useful in some clinical settings include tests of platelet aggregation, which measure the ability of platelets to adhere to one another in response to agonists like thrombin; and quantitative and qualitative tests of von Willebrand factor (vWF), which plays an important role in platelet adhesion to the extracellular matrix (Chapter 4). An older test, the bleeding time, is time-consuming and difficult to standardize and has been discarded. Instrument-based assays that provide quantitative measures of platelet function are used in some centers but remain imperfect at predicting bleeding risk, presumably because of difficulties in simulating in vivo clotting in the laboratory.

More specialized tests are available to measure the levels of specific clotting factors, fibrinogen, fibrin split products, and the presence of circulating anticoagulants.

Bleeding Disorders Caused by Vessel Wall Abnormalities

Disorders in this category are relatively common but do not usually cause serious bleeding problems. Most often, they present with small hemorrhages (petechiae and purpura) in the skin or mucous membranes, particularly the gingivae. On occasion, more significant hemorrhages occur into joints, muscles, and subperiosteal locations, or take the form of menorrhagia, nosebleeds, gastrointestinal bleeding, or hematuria. The platelet count and tests of coagulation (PT, PTT) are usually normal, pointing by exclusion to the underlying problem.

The clinical conditions in which vessel wall abnormalities cause bleeding include:

- *Infections* often induce petechial and purpuric hemorrhages, particularly meningococcemia, other forms of septicemia, infective endocarditis, and several of the rickettsioses. The involved mechanisms include microbial damage to the microvasculature (vasculitis) and disseminated intravascular coagulation.
- *Drug reactions* sometimes take the form of cutaneous petechiae and purpura without causing thrombocytopenia. In many instances the vascular injury is mediated by the deposition of drug-induced immune complexes in vessel walls, leading to hypersensitivity (*leukocytoclastic*) vasculitis (Chapter 11).
- *Scurvy* and *Ehlers-Danlos syndrome* are associated with microvascular bleeding due to collagen defects that weaken vessel walls. Acquired vascular fragility accounts for the spontaneous purpura that are commonly seen in older adults and the skin hemorrhages that are seen with *Cushing syndrome*, in which the protein-wasting effects of excessive corticosteroid production cause loss of perivascular extracellular matrix.
- *Henoch-Schönlein purpura* is a systemic immune disorder characterized by purpura, colicky abdominal pain, polyarthralgia, and acute glomerulonephritis (Chapter 20). These changes result from the deposition of circulating immune complexes within vessels throughout the body and within the glomerular mesangial regions.
- *Hereditary hemorrhagic telangiectasia* (also known as *Rendu-Osler-Weber syndrome*) is an autosomal dominant disorder that can be caused by mutations in at least five different genes, most of which modulate TGF- β signaling. It is characterized by dilated, tortuous blood vessels with thin walls that bleed readily. Bleeding can occur anywhere, but it is most common under the mucous membranes of the nose (epistaxis), tongue, mouth, and eyes, and throughout the gastrointestinal tract.
- *Perivascular amyloidosis* can weaken blood vessel walls and cause bleeding. This complication is most common with amyloid light-chain (AL) amyloidosis (Chapter 6) and often manifests as mucocutaneous petechiae.

Among these conditions, serious bleeding is most often associated with hereditary hemorrhagic telangiectasia. The bleeding in each is nonspecific, and the diagnosis is based on the recognition of other more specific associated findings.

Bleeding Related to Reduced Platelet Number: Thrombocytopenia

Reduction in platelet number (thrombocytopenia) constitutes an important cause of generalized bleeding. A count less than 150,000 platelets per μL is generally considered to constitute thrombocytopenia. Platelet counts in the range of 20,000 to 50,000 platelets per μL can aggravate posttraumatic bleeding, and platelet counts less than 20,000 platelets per μL may be associated with spontaneous (nontraumatic) bleeding. When thrombocytopenia is isolated, the PT and PTT are normal.

You will recall that following a vascular injury, platelets initially adhere and aggregate to form the primary hemostatic plug and then promote key reactions in the coagulation cascade that lead to secondary hemostasis and formation of a fibrin clot (Chapter 4). Spontaneous bleeding associated with thrombocytopenia most often involves small vessels. Common sites for such hemorrhages are the skin and the mucous membranes of the gastrointestinal and genitourinary tracts. Most feared, however, is intracranial bleeding, which is a threat to any patient with a markedly depressed platelet count.

TABLE 14.9 Causes of Thrombocytopenia

Decreased Production of Platelets
Selective impairment of platelet production Drug-induced: alcohol, thiazides, cytotoxic drugs Infections: measles, human immunodeficiency virus (HIV) Nutritional deficiencies: B ₁₂ , folate deficiency (megaloblastic leukemia) Aplastic anemia (see Table 14.7) Bone marrow replacement Leukemia, disseminated cancer, granulomatous disease Ineffective hematopoiesis Myelodysplastic syndromes (Chapter 13)
Decreased Survival of Platelets
Immunologic Destruction
Chronic immune thrombocytopenic purpura Acute immune thrombocytopenic purpura Systemic lupus erythematosus, B-cell lymphoid neoplasms Alloimmune: posttransfusion and neonatal Drug-associated: quinidine, heparin, sulfa compounds Infections: HIV, infectious mononucleosis (transient, mild), dengue fever
Nonimmunologic Destruction
Disseminated intravascular coagulation Thrombotic microangiopathies Giant hemangiomas
Sequestration
Hypersplenism
Dilution
Transfusions

The causes of thrombocytopenia fall into four major categories (Table 14.9).

- *Decreased platelet production.* This can result from conditions that depress marrow output generally (such as aplastic anemia and leukemia) or affect megakaryocytes selectively. Examples of the latter include certain drugs and alcohol, which may suppress platelet production through uncertain mechanisms when taken in large amounts; HIV, which may infect megakaryocytes and inhibit platelet production; and myelodysplastic neoplasm (Chapter 13), which occasionally presents with isolated thrombocytopenia.
- *Decreased platelet survival.* This may have an immunologic or non-immunologic basis. In immune thrombocytopenia, destruction is caused by the deposition of antibodies or immune complexes on platelets. Autoimmune thrombocytopenia is discussed in the following section. Alloimmune thrombocytopenia can arise when platelets are transfused or when platelets cross the placenta from the fetus into the pregnant mother. In the latter case, IgG antibodies made in the mother may cause clinically significant thrombocytopenia in the fetus. This is reminiscent of hemolytic disease of the newborn, in which red cells are the target (Chapter 10). The most important nonimmunologic causes are *disseminated intravascular coagulation (DIC)* and the *thrombotic microangiopathies*, in which unbridled, often systemic, platelet consumption reduces platelet life span. Non-immunologic destruction of platelets may also be caused by mechanical injury, such as in individuals with prosthetic heart valves.
- *Sequestration.* The spleen normally sequesters 30% to 35% of the body's platelets, but this can rise to 80% to 90% when the spleen is enlarged, producing moderate degrees of thrombocytopenia.
- *Dilution.* Massive transfusions can produce dilutional thrombocytopenia.

Chronic Immune Thrombocytopenic Purpura

Chronic immune thrombocytopenic purpura (ITP) is caused by autoantibody-mediated destruction of platelets. It can occur in the setting of a variety of predisposing conditions and exposures (secondary) or in the absence of any known risk factors (primary or idiopathic). The contexts in which chronic ITP occurs secondarily are numerous and include individuals with systemic lupus erythematosus (Chapter 6), HIV infection, and B-cell neoplasms such as chronic lymphocytic leukemia (Chapter 13). The diagnosis of primary chronic ITP is made only after secondary causes are excluded.

Pathogenesis

Autoantibodies, most often directed against platelet membrane glycoproteins IIb/IIIa or Ib/IX, can be demonstrated in the plasma and bound to the platelet surface in about 80% of patients. In most cases the antiplatelet antibodies are of the IgG class. As in autoimmune hemolytic anemias, antiplatelet antibodies act as opsonins that are recognized by IgG Fc receptors expressed on phagocytes (Chapter 6), leading to increased platelet destruction. The thrombocytopenia is usually markedly improved by splenectomy, indicating that the spleen is the major site of removal of opsonized platelets. The splenic red pulp is also rich in plasma cells, and part of the benefit of splenectomy may stem from the removal of a source of autoantibodies. In some instances, the autoantibodies may also bind to and damage megakaryocytes, leading to decreases in platelet production that further exacerbate the thrombocytopenia.

MORPHOLOGY

The principal changes of thrombocytopenic purpura are found in the spleen, bone marrow, and blood, but they are not specific. Secondary changes related to the bleeding diathesis may be found anywhere in the body. The spleen is of normal size. Typically, there is congestion of the sinusoids and enlargement of the splenic follicles, often associated with prominent reactive germinal centers. In many instances scattered megakaryocytes are found within the sinusoids, possibly representing a mild form of extramedullary hematopoiesis driven by elevated levels of thrombopoietin. **The marrow reveals a modestly increased number of megakaryocytes.** Some are apparently immature, with large, nonlobulated, single nuclei. These findings are not specific but merely reflect accelerated thrombopoiesis, being found in most forms of thrombocytopenia resulting from increased platelet destruction. The importance of bone marrow examination is to rule out thrombocytopenia due to marrow failure or other primary marrow disorders. The secondary changes relate to hemorrhage, often in the form of **petechial bleeds into the skin and mucous membranes.** The **peripheral blood often reveals unusually large platelets** (megathrombocytes), which are a sign of accelerated thrombopoiesis.

Clinical Features

Chronic ITP occurs most commonly in adult females younger than 40 years of age. The female-to-male ratio is 3:1. It is often insidious in onset and is characterized by bleeding into the skin and mucosal surfaces. Pinpoint hemorrhages (*petechiae*) are especially prominent in the dependent areas where the capillary pressure is higher. Petechiae can become confluent, giving rise to *ecchymoses*. Often there is a history of easy bruising, nosebleeds, gingival bleeding, and hemorrhages into soft tissues from relatively minor trauma. The disease may manifest first with melena, hematuria, or excessive menstrual flow. Subarachnoid hemorrhage and intracerebral hemorrhage are serious and sometimes fatal complications, but fortunately are rare in treated patients. Splenomegaly and lymphadenopathy are not seen in primary disease, and their presence should lead one to consider other diagnoses, such as ITP secondary to a B-cell neoplasm.

Typical laboratory findings are reflective of isolated thrombocytopenia. A low platelet count, normal or increased megakaryocytes in the bone marrow, and large platelets in the peripheral blood are taken as presumptive evidence of accelerated platelet destruction. The PT and PTT are normal. Tests for platelet autoantibodies have low sensitivity and specificity and are not clinically useful. Therefore, the diagnosis is one of exclusion and can be made only after other causes of thrombocytopenia (such as those listed in Table 14.9) have been ruled out.

Almost all patients respond to glucocorticoids (which inhibit phagocyte function), but many relapse following withdrawal of steroids. Those with moderately severe thrombocytopenia (platelet counts $>30,000$ per μL) can be followed carefully, and in some of these individuals ITP may spontaneously remit. In individuals with severe thrombocytopenia, splenectomy normalizes the platelet count in about two-thirds of patients, but with the attendant increased risk of bacterial sepsis. Immunomodulatory agents such as intravenous immunoglobulin or anti-CD20 antibody (rituximab) are often effective in patients who relapse after splenectomy or for whom splenectomy is contraindicated. Peptides and small molecules that mimic the effects of thrombopoietin (so-called *TPO-mimetics*) may also improve platelet counts in individuals with disease that is refractory to other treatments.

Acute Immune Thrombocytopenic Purpura

Like chronic ITP, this condition is caused by autoantibodies to platelets, but its clinical features and course are distinct. Acute ITP is mainly a disease of childhood occurring with equal frequency in both sexes. Symptoms appear abruptly, often 1 to 2 weeks after a self-limited viral illness, which appears to trigger the development of autoantibodies through uncertain mechanisms. Unlike chronic ITP, acute ITP is self-limited, usually resolving spontaneously within 6 months. Glucocorticoids are given only if the thrombocytopenia is severe. In about 20% of children, usually those without a viral prodrome, thrombocytopenia persists; these children have a childhood form of chronic ITP that follows a course similar to the adult disease.

Drug-Induced Thrombocytopenia

Drugs can induce thrombocytopenia through direct effects on platelets and secondary to immunologically mediated platelet destruction. The drugs most implicated are quinine, quinidine, and vancomycin, all of which induce drug-dependent antibody binding to platelet glycoproteins. Much more rarely, drugs induce true autoantibodies through unknown mechanisms. Thrombocytopenia, which may be severe, can occur in those who are taking platelet inhibitory drugs that bind glycoprotein IIb/IIIa; it is hypothesized that these drugs induce conformational changes in glycoprotein IIb/IIIa and create an immunogenic epitope.

Heparin-induced thrombocytopenia (HIT) has a distinctive pathogenesis and is of particular importance because of its potential for severe clinical consequences (Chapter 4). Thrombocytopenia occurs in about 5% of persons receiving heparin and is of two types:

- *Type I HIT occurs rapidly after the onset of therapy and is of little clinical importance*, sometimes resolving despite the continuation of therapy. It most likely results from a direct platelet-aggregating effect of heparin.
- *Type II HIT is less common but is often life threatening.* It occurs 5 to 14 days after therapy begins (or sooner if the person has been sensitized to heparin) and, paradoxically, often leads to venous and arterial thrombosis. It is caused by antibodies that recognize complexes of heparin and platelet factor 4, a normal component of platelet granules. Binding of antibody to these complexes activates platelets and promotes thrombosis, even in the setting of thrombocytopenia. Failure to immediately stop heparin and institute an

alternative nonheparin anticoagulant may lead to clot formation within large arteries, vascular insufficiency, and limb loss, as well as deep venous thrombosis with the attendant risk of pulmonary thromboembolism. The risk of type II HIT is lowered but not eliminated by use of low-molecular-weight heparin preparations. Once type II HIT develops, even low-molecular-weight heparins exacerbate the thrombotic tendency and must be avoided.

HIV-Associated Thrombocytopenia

Thrombocytopenia is one of the most common hematologic manifestations of HIV infection. Both decreased platelet production and increased platelet destruction contribute. CD4 and CXCR4, the receptor and coreceptor, respectively, for HIV, are found on megakaryocytes, allowing these cells to be infected. HIV-infected megakaryocytes are prone to apoptosis and their ability to produce platelets is impaired. HIV infection also causes B-cell hyperplasia and dysregulation (possibly due to its direct effects on CD4⁺ T cells), which predisposes to the development of autoantibodies. In some instances, the antibodies are directed against platelet membrane glycoprotein IIb–IIIa complexes. As in other immune cytopenias, the autoantibodies opsonize platelets, promoting their destruction by mononuclear phagocytes in the spleen and elsewhere. The deposition of immune complexes on platelets may also contribute to thrombocytopenia in some HIV-infected patients.

Thrombotic Microangiopathies: Thrombotic Thrombocytopenic Purpura and Hemolytic Uremic Syndrome

Thrombotic microangiopathies encompass a spectrum of clinical syndromes that have in common abnormalities that lead to excessive activation of platelets, which form thrombi in small blood vessels. This group of disorders includes thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS).

According to its original description, TTP was defined as the pentad of fever, thrombocytopenia, microangiopathic hemolytic anemia, transient neurologic deficits, and renal failure. HUS is also associated with microangiopathic hemolytic anemia and thrombocytopenia but is distinguished by the absence of neurologic symptoms, the prominence of acute renal failure, and its frequent occurrence in children. With time, experience, and increased mechanistic insight, however, these distinctions have blurred. Many adult patients with “TTP” lack one or more of the five criteria, and some patients with “HUS” have fever and neurologic dysfunction.

In both conditions, intravascular thrombi cause a microangiopathic hemolytic anemia and widespread organ dysfunction, and the attendant consumption of platelets leads to thrombocytopenia. It is believed that the varied clinical manifestations of TTP and HUS are related to differing proclivities for thrombus formation in tissues. Although DIC (discussed later) and thrombotic microangiopathies share features such as microvascular occlusion and microangiopathic hemolytic anemia, they are pathogenically distinct. In TTP and HUS (unlike in DIC), activation of the coagulation cascade is not of primary importance, and hence laboratory tests of coagulation, such as the PT and PTT, are usually normal.

Although certain features of the various thrombotic microangiopathies overlap, the triggers for the pathogenic platelet activation are distinctive and provide a more satisfying and clinically relevant way of thinking about these disorders (summarized in [Table 14.10](#)). **TTP is caused by a deficiency in a plasma enzyme called ADAMTS13, also designated “vWF metalloprotease.” ADAMTS13 is a metalloprotease that degrades very high-molecular-weight multimers of vWF.** In its absence, large multimers accumulate in plasma and tend to promote spontaneous platelet activation and aggregation. Superimposition of

TABLE 14.10 Thrombotic Microangiopathies: Causes and Associations

Thrombotic Thrombocytopenic Purpura
Deficiency of ADAMTS13
Inherited
Acquired (autoantibodies)
Hemolytic Uremic Syndrome
Typical
<i>Escherichia coli</i> strain O157:H7 infection
Endothelial damage by Shiga-like toxin
Atypical
Alternative complement pathway inhibitor deficiencies (complement factor H, membrane cofactor protein, or factor I)
Inherited
Acquired (autoantibodies)
Miscellaneous Associations
Drugs (cyclosporine, chemotherapeutic agents)
Radiation, bone marrow transplantation
Other infections (HIV, pneumococcal sepsis)
Conditions associated with autoimmunity (systemic lupus erythematosus, HIV infection, lymphoid neoplasms)

HIV, Human immunodeficiency virus.

endothelial cell injury (caused by some other condition) may further promote the formation of platelet aggregates, thus initiating or exacerbating clinically evident TTP.

ADAMTS13 deficiency may be inherited or acquired. In the acquired form, an autoantibody that inhibits the metalloprotease activity of ADAMTS13 is present. Less commonly, patients inherit an inactivating mutation in *ADAMTS13*. In those with hereditary ADAMTS13 deficiency, the onset of TTP is often delayed until adolescence, and the symptoms are episodic. Thus, factors other than ADAMTS13 deficiency (e.g., a superimposed vascular injury or prothrombotic state) must be involved in triggering full-blown TTP.

TTP is an important diagnosis to consider in any patient presenting with thrombocytopenia and microangiopathic hemolytic anemia because delays in diagnosis can be fatal. With plasma exchange, which removes autoantibodies and provides functional ADAMTS13, TTP (once uniformly fatal) can be treated successfully in more than 80% of patients.

By contrast, HUS is associated with normal levels of ADAMTS13 and is initiated by several other distinct defects. It occurs in hereditary and acquired forms:

- **Hereditary forms of HUS are caused by mutations that lead to excessive activation of complement.** Affected genes encode proteins such as complement factor 3 (C3), CD46, and complement factors B, H, and I, all of which are components or regulators of the alternative complement pathway. This form of HUS tends to be episodic and relapsing. It is proposed that some triggering event such as infection in a susceptible individual leads to uncontrolled complement activation.
- **Acquired forms of HUS may be triggered by diverse insults, including specific types of infections, autoantibodies, and toxic exposures.** The most common infectious cause is gastroenteritis due to *Escherichia coli* strain O157:H7, which elaborates a Shiga-like toxin. This toxin is absorbed from the inflamed gastrointestinal mucosa into the circulation, but precisely how it triggers HUS is unknown. Children and older adults are at highest risk. Those affected present with bloody diarrhea, and a few days later HUS makes its appearance. Other acquired causes include inhibitory autoantibodies

against complement regulatory factors and exposure to certain drugs, including some chemotherapeutic agents.

Unlike TTP, the basis for the platelet activation in hereditary and acquired forms of HUS is unclear. As in paroxysmal nocturnal hemoglobinuria (discussed earlier), therapeutic antibodies that inhibit the activation of the complement factor C5 are effective in preventing thrombosis in patients with hereditary HUS, proving that excessive complement activation has a central role in this form of the disease. Immunosuppression may be beneficial to patients with inhibitory autoantibodies. HUS associated with infection is treated supportively; those who survive the acute insult usually recover completely, but some have permanent renal damage and require dialysis or renal transplantation. The impact of HUS and TTP on the kidneys is discussed further in Chapter 20.

Thrombotic microangiopathies resembling HUS may also be seen following exposures to other agents that damage endothelial cells (e.g., certain drugs and radiation therapy). The prognosis in these settings is guarded because the HUS is often complicated by chronic, life-threatening conditions.

Bleeding Disorders Related to Defective Platelet Function

Qualitative defects of platelet function can be inherited or acquired. Several inherited disorders characterized by abnormal platelet function and normal platelet counts have been described. A brief discussion of these rare diseases is warranted because they provide insight into the molecular mechanisms of platelet function.

Inherited disorders of platelet function can be classified into three pathogenically distinct groups: (1) defects of adhesion, (2) defects of aggregation, and (3) disorders of platelet secretion (release reaction).

- *Bernard-Soulier syndrome* illustrates the consequences of defective adhesion of platelets to subendothelial matrix. It is an autosomal recessive disorder caused by an inherited deficiency of the platelet membrane glycoprotein complex Ib/IX. This glycoprotein is a receptor for vWF and is essential for normal platelet adhesion to the subendothelial extracellular matrix (Chapter 4). Affected patients have a variable, often severe, bleeding tendency.
- *Glanzmann thrombasthenia* exemplifies bleeding due to defective platelet aggregation. It is transmitted as an autosomal recessive trait. Thrombasthenic platelets fail to aggregate in response to adenosine diphosphate (ADP), collagen, epinephrine, or thrombin because of deficiency or dysfunction of glycoprotein IIb/IIIa, an integrin that participates in “bridge formation” between platelets by binding fibrinogen. The associated bleeding tendency is often severe.
- *Storage pool disorders* are characterized by the defective release of certain mediators of platelet activation, such as thromboxanes and granule-bound ADP. The biochemical defects underlying these disorders are varied, complex, and beyond the scope of our discussion.

Among the acquired defects of platelet function, two are clinically significant. The first is caused by ingestion of aspirin and other nonsteroidal antiinflammatory drugs. Aspirin is a potent, irreversible inhibitor of cyclooxygenase, an enzyme that is required for the synthesis of thromboxane A₂ and prostaglandins (Chapter 3). These mediators play important roles in platelet aggregation and subsequent release reactions (Chapter 4). The antiplatelet effects of aspirin form the basis for its use in preventing coronary thrombosis (Chapter 12). Uremia (Chapter 20) is the second common condition associated with acquired defects in platelet function. The pathogenesis of platelet dysfunction in uremia is complex and involves defects in adhesion, granule secretion, and aggregation.

Bleeding Disorders Related to Abnormalities in Clotting Factors

Inherited or acquired deficiencies of virtually every coagulation factor have been reported as causes of abnormal bleeding. Bleeding due to coagulation factor deficiencies commonly manifest as large posttraumatic ecchymoses or hematomas, or prolonged bleeding from a laceration or after a surgical procedure. Unlike bleeding seen with thrombocytopenia, bleeding due to coagulation factor deficiencies often occurs in the gastrointestinal and urinary tracts and in weight-bearing joints (hemarthrosis). Typical stories include the patient who oozes blood for days after a tooth extraction or who develops hemarthrosis after minor stress on a knee joint.

Hereditary deficiencies typically affect a single clotting factor. The most common and important inherited deficiencies of coagulation factors affect factor VIII (*hemophilia A*) and factor IX (*hemophilia B*). Deficiencies of vWF (*von Willebrand disease*) are also discussed here, as vWF is involved in both coagulation and platelet function.

Acquired deficiencies usually involve multiple coagulation factors and can be caused by decreased protein synthesis or a shortened protein half-life. Vitamin K deficiency (Chapter 9) impairs the synthesis of factors II, VII, IX, X and protein C. Many coagulation factors are made in the liver, and inadequate synthesis is often observed in severe parenchymal liver disease. By contrast, in DIC, multiple coagulation factors are consumed, leading to their deficiency. Acquired deficiencies of single factors occur but are rare; these are usually caused by inhibitory autoantibodies.

Factor VIII–vWF Complex

The two most common inherited disorders of bleeding, hemophilia A and von Willebrand disease, are caused by qualitative or quantitative defects involving factor VIII and vWF, respectively. Before we discuss these disorders, it will be helpful to review the structure and function of these two proteins, which exist together in the plasma as part of a single large complex.

Factor VIII is an essential cofactor of factor IX, which converts factor X to factor Xa (Fig. 14.25; Chapter 4). It is made by endothelial cells, whereas vWF is made by both endothelial cells and megakaryocytes, which are the source of the vWF that is present in platelet α -granules. Once secreted into the blood, factor VIII binds to and is stabilized by vWF; an interaction that increases the half-life of factor VIII from about 2.4 hours to about 12 hours.

vWF secreted into the circulation by endothelial cells exists as multimers containing as many as 100 subunits that can exceed 20×10^6 Daltons in molecular mass. Some of the secreted vWF is also deposited in the subendothelial matrix, where it lies ready to promote platelet adhesion if the endothelial lining is disrupted (see Fig. 14.25). In addition to factor VIII, vWF interacts with several other proteins involved in hemostasis, including collagen, heparin, and platelet membrane glycoproteins. The most important hemostatic function of vWF is to promote the adhesion of platelets to the subendothelial matrix. This occurs through bridging interactions between platelet glycoprotein Ib/IX, vWF, and matrix components such as collagen. vWF may also promote platelet aggregation by binding to activated glycoprotein IIb/IIIa integrins, an activity that may be of particular importance under conditions of high shear stress (such as occurs in small vessels).

Factor VIII and vWF protein levels are measured by immunologic techniques. Factor VIII function is measured by performing coagulation assays with mixtures of patient plasma and factor VIII-deficient plasma, and vWF function is assessed using the ristocetin agglutination test. The latter is performed by adding patient plasma to normal formalin-fixed platelets in the presence of ristocetin, a small molecule

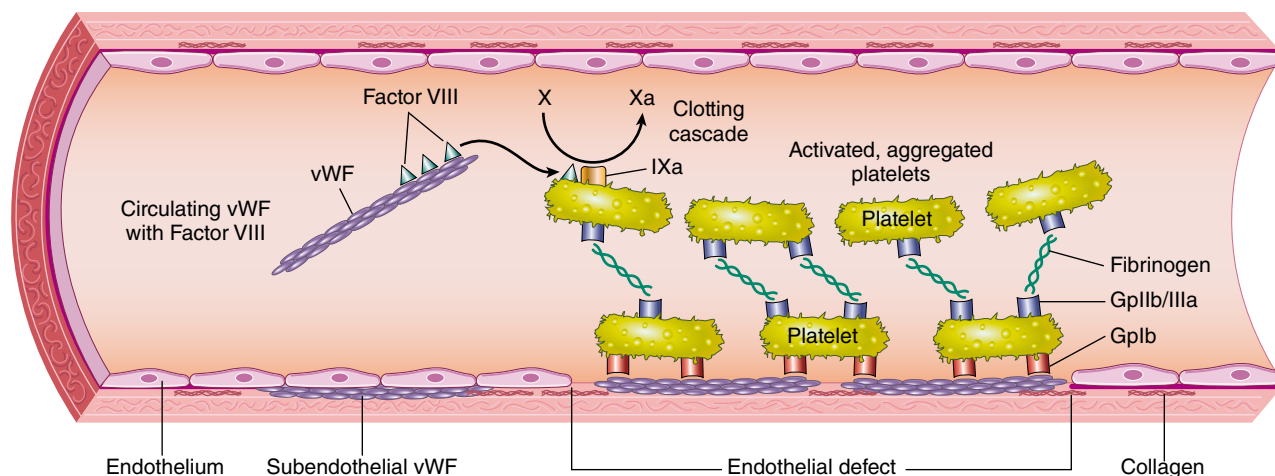


Fig. 14.25 Structure and function of factor VIII–von Willebrand factor (vWF) complex. Factor VIII is synthesized in the liver and kidney, and vWF is made in endothelial cells and megakaryocytes. The two associate to form a complex in the circulation. vWF is also present in the subendothelial matrix of normal blood vessels and the α -granules of platelets. Following endothelial injury, exposure of subendothelial vWF causes adhesion of platelets, primarily via the glycoprotein Ib (GpIb) platelet receptor. Circulating vWF and vWF released from the α -granules of activated platelets can bind exposed subendothelial matrix, further contributing to platelet adhesion and activation. Activated platelets form hemostatic aggregates; fibrinogen participates in aggregation through bridging interactions with the glycoprotein IIb/IIIa (GpIIb/IIIa) platelet receptor. Factor VIII takes part in the coagulation cascade as a cofactor in the activation of factor X on the surface of activated platelets.

that binds and “activates” vWF. Ristocetin induces multivalent vWF multimers to bind platelet glycoprotein Ib/IX and form interplatelet “bridges.” The resultant clumping (agglutination) of platelets reflects the vWF activity of the sample.

von Willebrand Disease

von Willebrand disease is the most common inherited bleeding disorder of humans, affecting about 1% of adults in the United States. The bleeding tendency is usually mild and often goes unnoticed until some hemostatic stress, such as surgery or a dental procedure, reveals its presence. The most common presenting symptoms are spontaneous bleeding from mucous membranes (e.g., epistaxis), excessive bleeding from wounds, or menorrhagia. It is usually transmitted as an autosomal dominant disorder, but rare autosomal recessive variants also exist.

von Willebrand disease is clinically and molecularly heterogeneous; hundreds of vWF variants have been described, only a few of which have been formally proven to cause disease. Three types of von Willebrand disease are recognized, each with a range of phenotypes:

- *Type 1 and type 3 von Willebrand disease are associated with quantitative defects in vWF.* Type 1, an autosomal dominant disorder characterized by a mild to moderate vWF deficiency, accounts for about 70% of all cases. Incomplete penetrance and variable expressivity are commonly observed, but it generally is associated with mild disease. Type 1 disease is caused by a spectrum of mutations, including point substitutions that interfere with maturation of the vWF protein or result in its rapid clearance from the plasma. Type 3 disease is an autosomal recessive disorder usually caused by deletions or frameshift mutations involving both alleles, resulting in little to no vWF synthesis. Because vWF stabilizes factor VIII in the circulation, factor VIII levels are also reduced in type 3 disease; as a result, the associated bleeding disorder is often severe.
- *Type 2 von Willebrand disease is characterized by qualitative defects in vWF;* there are several subtypes, of which type 2A is the most

common. It is inherited as an autosomal dominant disorder. vWF is expressed in normal amounts, but missense mutations are present that lead to defective multimer assembly. As a result, large- and intermediate-sized multimers, the most active forms of vWF, are missing from plasma. Type 2 von Willebrand disease accounts for 25% of all cases and is associated with mild to moderate bleeding.

Patients with von Willebrand disease have defects in platelet function despite having normal platelet counts. The plasma level of active vWF, measured as the ristocetin cofactor activity, is reduced. Because a deficiency of vWF decreases the stability of factor VIII, type 1 and type 3 von Willebrand disease is associated with a prolonged PTT.

Even within families in which there is a single defective vWF allele, wide variability in clinical expression is common. This is due in part to modifying genes that influence circulating levels of vWF, which have a broad range in normal populations. Persons with types 1 or 2 von Willebrand disease facing hemostatic challenges (dental work, surgery) can be treated with *desmopressin* (which stimulates vWF release) or infusions of plasma concentrates containing factor VIII and vWF or recombinant vWF. By contrast, rare patients with type 3 disease must be treated prophylactically with plasma concentrates and factor VIII infusions to prevent severe hemophilia-like bleeding.

Hemophilia A (Factor VIII Deficiency)

Hemophilia A, the most common hereditary disease associated with life-threatening bleeding, is caused by mutations in the gene encoding factor VIII, an essential cofactor for factor IX in the coagulation cascade. Hemophilia A is inherited as an X-linked recessive trait and thus affects mainly males and homozygous females. Rarely, excessive bleeding occurs in heterozygous females, presumably due to inactivation of the X chromosome bearing the normal factor VIII allele by chance in most cells (unfavorable lyonization). About 30% of patients have no family history; their disease is caused by new mutations.

Hemophilia A exhibits a wide range of clinical severity that correlates well with the level of factor VIII activity. Those with less than 1% of normal levels have severe disease; those with 2% to 5% of normal levels have moderately severe disease; and those with 6% to 50% of normal levels have mild disease. The varying degrees of factor VIII deficiency are largely explained by heterogeneity in the causative mutations. The most severe deficiencies result from an inversion involving the X chromosome that completely abolishes the synthesis of factor VIII. Less commonly, severe hemophilia A is associated with point mutations in factor VIII that impair the function of the protein. In such cases, factor VIII protein levels may be normal by immunoassay. Mutations permitting some active factor VIII to be synthesized are associated with mild to moderate disease.

Clinically, there is a tendency toward easy bruising and massive hemorrhage after trauma or surgery. In addition, “spontaneous” hemorrhages frequently occur in regions of the body that are susceptible to trauma, particularly the joints, where they are known as *hemarthroses*. Recurrent bleeding into the joints leads to progressive deformities that can be disabling. Petechiae are characteristically absent.

Patients with hemophilia A have a prolonged PTT and a normal PT, results that point to an abnormality of the intrinsic coagulation pathway. Factor VIII–specific assays are required for diagnosis. As explained in Chapter 4, the bleeding diathesis reflects the preeminent role of the factor VIIIa/factor IXa complex in activation of factor X *in vivo*. The precise explanation for the tendency of hemophiliacs to bleed at particular sites (joints, muscles, and the central nervous system) remains uncertain.

Hemophilia A may be treated with infusions of recombinant factor VIII or with a bispecific antibody that links factor IXa and factor Xa, thereby bypassing the need for factor VIII. An advantage of the bispecific antibody is that it remains active in patients who develop antibodies that bind and inhibit factor VIII, which occurs in about 15% of patients with severe hemophilia A following treatment with recombinant factor VIII. A disadvantage of both these therapies is their very high cost, which makes treatment of patients in low-resource countries very challenging.

Hemophilia B (Christmas Disease, Factor IX Deficiency)

Severe factor IX deficiency produces a disorder clinically indistinguishable from factor VIII deficiency (hemophilia A). This should not be surprising, given that factors VIII and IX function together to activate factor X. A wide spectrum of causative mutations involving the gene that encodes factor IX are found in patients with hemophilia B. Like hemophilia A, it is inherited as an X-linked recessive trait and shows variable clinical severity. In about 15% of patients, factor IX protein is present but is nonfunctional. As with hemophilia A, the PTT is prolonged and the PT is normal. Diagnosis of Christmas disease (named after the first patient identified with this condition, not the holiday) is made by assay of factor IX levels. Historically, the disease has been treated with infusions of recombinant factor IX. Recently, a gene therapy for this disorder has been approved that consists of a one-time infusion of a recombinant virus that delivers the coding sequence for factor IX to hepatocytes.

Disseminated Intravascular Coagulation

Disseminated intravascular coagulation (DIC) is an acute, subacute, or chronic thrombohemorrhagic disorder characterized by excessive activation of coagulation factors and the formation of thrombi in the microvasculature. It occurs as a secondary complication of many disorders. Sometimes the coagulopathy is localized to a specific organ or tissue. The thrombotic diathesis leads to the consumption of platelets, fibrin, and coagulation factors and, secondarily,

activation of fibrinolysis. DIC can present with signs and symptoms relating to tissue hypoxia and infarction caused by microthrombi; hemorrhage due to depletion of factors required for hemostasis and activation of fibrinolytic mechanisms; or both.

Etiology and Pathogenesis

It must be emphasized that DIC is not a primary disease, but rather is an acquired coagulopathy that may occur in the course of a variety of clinical conditions. In discussing the general mechanisms underlying DIC, it is useful to briefly review the normal process of blood coagulation and clot removal (Chapter 4).

Clotting *in vivo* is initiated by exposure of tissue factor, which activates factor VII. The most important effect of tissue factor/factor VII complexes is activation of factor IX, which in turn activates factor X. Activation of factor X leads to the generation of thrombin, the central player in clotting. At sites where the endothelium is disrupted, thrombin converts fibrinogen to fibrin; feeds back to activate factors IX, VIII, and V; stimulates fibrin cross-linking; inhibits fibrinolysis; and activates platelets, all of which augment the formation of a stable clot. To prevent runaway clotting, this process must be sharply limited to the site of tissue injury. Remarkably, as thrombin is swept away in the bloodstream and encounters uninjured vessels, it is converted to an anticoagulant through binding to thrombomodulin, a protein found on the surface of endothelial cells. The thrombin-thrombomodulin complex activates protein C, an important inhibitor of factor V and factor VIII. Other activated coagulation factors are removed from the circulation by the liver, and, as you will recall, the blood also contains several potent fibrinolytic factors, such as plasmin. These and additional checks and balances normally ensure that just enough clotting occurs at the right place and time.

From this brief review, it should be clear that DIC may result from pathologic activation of coagulation or impairment of clot-inhibiting mechanisms. Because the latter rarely constitute primary mechanisms of DIC, we will focus on abnormal initiation of clotting.

Two major mechanisms trigger DIC: (1) release of tissue factor or other procoagulants into the circulation, and (2) widespread injury of endothelial cells. Procoagulants such as tissue factor can be derived from a variety of sources, such as the placenta in obstetric complications or tissues injured by trauma or burns. Mucus released from certain adenocarcinomas may also act as a procoagulant by directly activating factor X.

Endothelial injury can initiate DIC in several ways. Injuries that cause endothelial cell necrosis expose the subendothelial matrix, leading to the activation of platelets and the coagulation pathway. However, even subtle endothelial injuries can unleash procoagulant activity. One mediator of endothelial injury is TNF, which is implicated in DIC occurring with sepsis. TNF induces endothelial cells to express tissue factor on their cell surfaces and to decrease the expression of thrombomodulin, tilting the checks and balances that govern hemostasis toward coagulation. In addition, TNF up-regulates the expression of adhesion molecules on endothelial cells, thereby promoting the adhesion of leukocytes, which can damage endothelial cells by releasing reactive oxygen species and preformed proteases. Widespread endothelial injury may also be produced by deposition of antigen-antibody complexes (e.g., systemic lupus erythematosus), temperature extremes (e.g., heat stroke, burns), or microorganisms (e.g., meningococci, rickettsiae).

DIC is most associated with obstetric complications, malignant neoplasms, sepsis, and major trauma. The triggers in these conditions are often multiple and interrelated. For example, in bacterial infections, endotoxins can inhibit the endothelial expression of thrombomodulin directly or indirectly by stimulating immune cells to make TNF and can also activate factor XII. Antigen-antibody complexes

formed in response to infection can activate the classic complement pathway, giving rise to complement fragments that secondarily activate platelets and granulocytes. In massive trauma, extensive surgery, and severe burns, the major trigger is the release of procoagulants such as tissue factor. In obstetric conditions, procoagulants derived from the placenta, dead retained fetus, or amniotic fluid may enter the circulation. Hypoxia, acidosis, and shock, which often coexist in very ill patients, may also cause widespread endothelial injury, and supervening infections can complicate problems further. Among cancers, acute promyelocytic leukemia and adenocarcinomas of the lung, pancreas, colon, and stomach are most frequently associated with DIC.

The possible consequences of DIC are twofold (Fig. 14.26).

- **Widespread deposition of fibrin** within the microcirculation. This leads to ischemia of the more severely affected or more vulnerable organs and microangiopathic hemolytic anemia, which results from the fragmentation of red cells as they squeeze through the narrowed microvasculature.
- **Consumption of platelets and clotting factors** and the activation of plasminogen, leading to a hemorrhagic diathesis. Plasmin not only cleaves fibrin, but it also digests factors V and VIII, reducing their concentration further. In addition, fibrin degradation products resulting from fibrinolysis inhibit platelet aggregation, fibrin polymerization, and thrombin.

MORPHOLOGY

Thrombi are most often found in the brain, heart, lungs, kidneys, adrenals, spleen, and liver, in decreasing order of frequency, but any tissue can be affected. Affected kidneys may have small thrombi in the glomeruli that evoke

only reactive swelling of endothelial cells or, in severe cases, microinfarcts or even **bilateral renal cortical necrosis**. Numerous fibrin thrombi may be found in alveolar capillaries, sometimes associated with pulmonary edema and fibrin exudation, creating "**hyaline membranes**" reminiscent of acute respiratory distress syndrome (Chapter 15). In the central nervous system, fibrin thrombi may cause microinfarcts, occasionally complicated by simultaneous hemorrhage, which can sometimes lead to variable neurologic signs and symptoms. In meningococcemia, fibrin thrombi within the microcirculation of the adrenal cortex are the probable basis for the massive adrenal hemorrhages seen in **Waterhouse-Friderichsen syndrome** (Chapter 24). An unusual form of DIC occurs in association with giant hemangiomas (**Kasabach-Merritt syndrome**) in which thrombi form within the neoplasm because of stasis and recurrent trauma to fragile blood vessels.

Clinical Features

The onset of DIC may be fulminant, as in sepsis or amniotic fluid embolism, or insidious and chronic, as in cases of carcinomatosis or retention of a dead fetus. It is almost impossible to detail all the potential clinical presentations, but a few common patterns are worthy of description. These include microangiopathic hemolytic anemia; dyspnea, cyanosis, and respiratory failure; convulsions and coma; oliguria and acute renal failure; and sudden or progressive circulatory failure and shock. In general, acute DIC associated (for example) with obstetric complications or major trauma is dominated by a bleeding diathesis, whereas chronic DIC such as occurs in cancer patients tends to present with thrombotic complications. Diagnosis is based on clinical observation and laboratory studies, including measurement of fibrinogen levels, platelets, PT and PTT, and fibrin degradation products, particularly D-dimers.

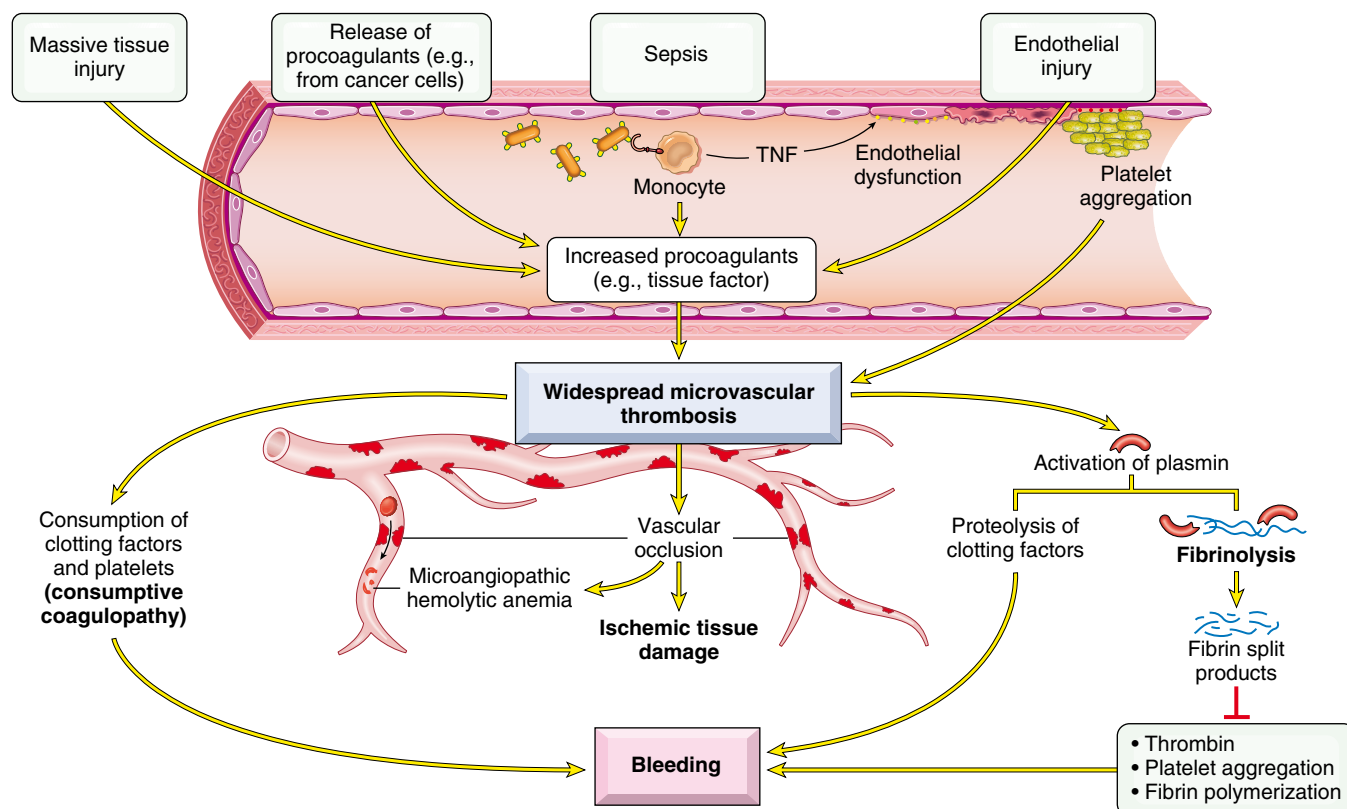


Fig. 14.26 Pathophysiology of disseminated intravascular coagulation. *TNF*, Tumor necrosis factor.

The prognosis is highly variable and largely depends on the underlying disorder. The only definitive treatment is to remove or treat the inciting cause. Management requires meticulous maneuvering between the Scylla of thrombosis and the Charybdis of bleeding diathesis. Administration of anticoagulants or procoagulants has been advocated in specific settings, but not without controversy.

KEY CONCEPTS

Immune Thrombocytopenic Purpura

- Caused by autoantibodies against platelet antigens
- May be triggered by drugs, infections, or lymphomas, or may be idiopathic

Thrombotic Thrombocytopenic Purpura and Hemolytic Uremic Syndrome

- Both manifest with thrombocytopenia, microangiopathic hemolytic anemia, and renal failure; fever and central nervous system involvement are more typical of TTP.
- TTP: Caused by acquired or inherited deficiencies of ADAMTS13, a plasma metalloprotease that cleaves very-high-molecular-weight multimers of vWF. Deficiency of ADAMTS13 results in abnormally large vWF multimers that activate platelets.
- HUS: Caused by inherited deficiencies of complement regulatory proteins or acquired abnormalities due to infection with Shiga-like toxin producing *E. coli* strain O157:H7, autoantibodies against complement factors, or exposure to certain drugs or radiation. These abnormalities initiate platelet activation, platelet aggregation, and microvascular thrombosis.

von Willebrand Disease

- Autosomal dominant disorder caused by mutations in vWF, a large protein that promotes the adhesion of platelets to subendothelial collagen and stabilizes factor VIII
- Typically causes a mild to moderate bleeding disorder resembling that associated with thrombocytopenia

Hemophilia

- Hemophilia A: X-linked disorder caused by mutations in factor VIII. Affected males typically present with severe bleeding into soft tissues and joints and have a prolonged PTT.
- Hemophilia B: X-linked disorder caused by mutations in coagulation factor IX. It is clinically identical to hemophilia A.

Disseminated Intravascular Coagulation

- Syndrome in which systemic activation of the coagulation leads to consumption of coagulation factors and platelets
- Can produce bleeding, vascular occlusion, and tissue hypoxemia in various combinations
- Common triggers: sepsis, major trauma, certain cancers, obstetric complications

Complications of Transfusion

Blood products are often rightly called the gift of life, permitting people to survive traumatic injuries and procedures such as hematopoietic stem cell transplantation and complex surgical procedures that would otherwise prove fatal. Over 4.5 million red cell transfusions are given in US hospitals each year. Thanks to improved screening of donors, blood products (packed red blood cells, platelets, and fresh-frozen plasma) are safer than ever before.

Nevertheless, complications still occur. Most are minor and transient. The most common is referred to as a *febrile nonhemolytic transfusion reaction*, which takes the form of fever and chills, sometimes with mild dyspnea, within 6 hours of a transfusion of red cells or

platelets. These reactions are thought to be caused by inflammatory mediators released from donor leukocytes. The frequency of these reactions increases with the storage age of the product and is decreased by measures that limit donor leukocyte contamination. Symptoms respond to antipyretics and are short-lived.

Other transfusion reactions are uncommon or rare, but can have severe and sometimes fatal consequences, and therefore merit discussion.

Allergic Reactions

Severe, potentially fatal allergic reactions may occur when blood products containing certain antigens are given to previously sensitized recipients. These are most likely to occur in patients with IgA deficiency, which has a frequency of about 1 in 600 people. In this instance, the reaction is triggered by IgG antibodies that recognize IgA in the infused blood product. Fortunately, most patients with IgA deficiency do not develop such antibodies and these severe reactions are rare, occurring in 1 in 20,000 to 1 in 50,000 transfusions. Urticarial allergic reactions may be triggered by the presence of an allergen in the donated blood product that is recognized by IgE antibodies in the recipient. These are considerably more common, occurring in 1% to 3% of transfusions, but are generally mild. In most instances, symptoms respond to antihistamines and do not require discontinuation of the transfusion.

Hemolytic Reactions

Acute hemolytic reactions are usually caused by preformed IgM antibodies against donor red cells that fix complement. They most commonly stem from an error in patient identification or tube labeling that allows a patient to receive an ABO-incompatible unit of blood. Preexisting high-affinity “natural” IgM antibodies, usually against polysaccharide blood group antigens A or B, bind to red cells and rapidly induce complement-mediated lysis, intravascular hemolysis, and hemoglobinuria. Fever, shaking chills, and flank pain appear rapidly. The direct Coombs test is typically positive, unless all the donor red cells have lysed. The signs and symptoms are due to complement activation rather than intravascular hemolysis per se, as osmotic lysis of red cells (e.g., by mistakenly infusing red cells and 5% dextrose in water simultaneously) produces hemoglobinuria without any of the other symptoms of a hemolytic reaction. In severe cases, the process may rapidly progress to DIC, shock, acute renal failure, and occasionally death.

Delayed hemolytic reactions are caused by antibodies that recognize red cell antigens that the recipient was sensitized to previously, for example, through a prior blood transfusion. These are typically caused by IgG antibodies to foreign protein antigens and are associated with a positive direct Coombs test and laboratory features of hemolysis (e.g., low haptoglobin and elevated lactate dehydrogenase). The responsible antibodies typically produce extravascular hemolysis, but the severity varies according to the antigen. Antibodies to antigens such as Rh, Kell, and Kidd may induce severe and potentially fatal reactions, apparently because of macrophage activation and release of cytokines. By contrast, delayed hemolytic reactions produced by other antibodies typically are associated with relatively minor signs and symptoms.

Transfusion-Related Acute Lung Injury

Transfusion-related acute lung injury (TRALI) is a severe, frequently fatal complication in which factors in a transfused blood product trigger the activation of neutrophils in the lung microvasculature. The incidence of TRALI is low, probably less than 1 in 10,000 transfusions, but it may occur more frequently in patients with preexisting lung disease. Though its pathogenesis is incompletely understood, current models favor a “two-hit” hypothesis. The first is neutrophil sequestration

and priming in the microvasculature of the lung. It is postulated that endothelial cells are involved both in sequestration and priming; the former by up-regulation of adhesion molecules and the latter by release of cytokines. The second hit involves activation of primed neutrophils by a factor present in the transfused blood product.

A variety of factors have been implicated as “second hits,” but the leading candidates are antibodies in the transfused blood product that recognize antigens expressed on neutrophils. By far the most common antibodies associated with TRALI are those that bind major histocompatibility complex (MHC) antigens. These antibodies are often found in multiparous females, who generate such antibodies in response to foreign MHC antigens expressed by the fetus. In other cases, donor antibodies to neutrophil-specific antigens have been implicated as triggers.

Although TRALI has been associated with virtually all plasma-containing blood products, it is more likely to occur following transfusion of products containing high levels of donor antibody, such as fresh-frozen plasma and platelets. The presentation is dramatic, consisting of sudden-onset respiratory failure, during or soon after a transfusion. Diffuse bilateral pulmonary infiltrates that do not respond to diuretics are seen on chest imaging. Other associated findings include fever, hypotension, and hypoxemia. The treatment is largely supportive, and the outcome is guarded; mortality is 5% in uncomplicated cases and up to 67% in the severely ill. TRALI is important to recognize, because donor products that induce the complication in one patient are much more likely to do so in a second. Indeed, measures taken to exclude multiparous females from plasma donation have sharply reduced the incidence of TRALI.

Infectious Complications

Virtually any infectious agent can be transmitted through blood products, but bacterial and viral infections are the dominant culprits. Most bacterial infections are caused by skin flora, indicating that the contamination occurred at the time that the product was taken from the donor. Significant bacterial contamination (sufficient to produce symptoms) is much more common in platelet preparations than in red cell preparations, due in large part to the fact that platelets (unlike red cells) must be stored at room temperature, a condition favorable for bacterial growth. Rates of bacterial infection following platelet transfusion are as high as 1 in 5000, with infections secondary to red cell transfusions being several orders of magnitude less frequent. Many of the symptoms (fever, chills, hypotension) resemble those of hemolytic and nonhemolytic transfusion reactions, and it may be necessary to start broad-spectrum antibiotics prospectively in symptomatic patients while awaiting laboratory results.

Advances in donor selection, donor screening, and microbial testing have dramatically decreased the incidence of viral transmission by blood products. However, on rare occasions when the donor is acutely infected but the virus is not yet detectable with current nucleic acid testing technology, there can be transfusion-related transmission of viruses such as HIV, hepatitis C, and hepatitis B. Rates of transmission

of HIV, hepatitis C, and hepatitis B are estimated to be 1 in 1.5 million, 1 in 1.2 million, and 1 in 1 million units of blood, respectively. There also remains a low risk from other infections with agents such as West Nile virus, trypanosomes, and babesia.

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